NIHR | National Institute for Health and Care Research

Summary

Reference Number	NIHR306398
Lead Applicant	Professor Lindsay Bearne
Research Title	Developing equitable, personalised community rehabilitation interventions for people with peripheral artery disease.
Plain English Summary	**Why is this research needed?** Out-of-hospital rehabilitation (known as community rehabilitation) improves people's health and wellbeing and saves money. But community rehabilitation is not provided equally, and it is not known how best to treat people with some conditions or whether treatments are acceptable to patients. Improving equal access to high quality community rehabilitation is central to the government's NHS plan. One of the conditions that we don't know how to rehabilitate in the community is called peripheral artery disease. Peripheral artery disease is poor leg blood flow due to narrowed arteries. People from Black, African Caribbean, South Asian or Eastern European communities, those with diabetes or living in areas with socioeconomic deprivation are affected more by peripheral artery disease. Severe peripheral artery disease can cause constant leg pain, sores and gangrene in the toes. This is called chronic limb threatening ischaemia. Most people need their arteries unblocked in hospital and then need to self-manage their recovery at home. But there are no self-management programmes to support people with chronic limb threatening ischaemia to recover and live well at home. Some people with milder peripheral arterial disease experience leg pain on walking (called intermittent claudication). Walking frequently is the best way to improve this pain and reduce the risk of severe artery disease. It is hard for people to walk on their own when they are in pain. New ways are needed to help people walk regularly, improve their leg pain and do more activities. Many people with peripheral artery disease are not active enough to reap the health benefits. This increases their risk of getting other health problems. There are many ways to help people to become active. One way is to use a smartwatch that records how much activity is done and sends messages to encourage people to be more active. Smartwatches can be used separately or as part of an activity programme. But some older people find smartwatches difficult to understand and use. **Aims** To work with patients, practitioners and service planners to co-design ways to provide community rehabilitation equitably for everyone with peripheral artery disease and explore new ways to help people to be more active. People from groups who are most affected by peripheral artery disease and underrepresented communities will be included. **What will happen?** I will build a team to 1. summarise the evidence for the factors that influence self-management, interview patients and practitioners and co-design a self-management programme to help people with chronic limb threatening ischaemia live well at home. 2. look at the results from a study of an existing walking exercise programme for people with intermittent claudication and use this information to adapt this programme so it can be provided in the community. 3. interview patients and practitioners, co-design and user test a new treatment including a smartwatch to help everyone with peripheral artery disease be more active. 4. co-design a plan to implement these treatments in the NHS, if these ways are shown to be effective and acceptable. **How have patients and communities been involved?** Members of the public and patients from different backgrounds chose this topic and helped design this research. They helped write this summary. The public advisory group will meet every 6-months and be involved in co-design and workstream management. **Sharing this research** Patients, practitioners, service planners and voluntary and professionals' organisations (like Circulation Foundation) will co-design and help share our research progress and findings, so it is available in the NHS and for the public. This may include writing research summaries; giving talks; hosting meetings; designing infographics,

	animations, websites and social media.
Total Research Cost	

Our purpose for collecting this data is to communicate with you about your application and have the necessary information to evaluate you for a grant. The data we collect here is collected in the public interest. Information provided here may be subject to Freedom of Information requests.

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1. Application Summary Information

Host organisation (which will administer any award)
City St George's, University of London

Partner organisation	St George's Healthcare NHS Trust

Research Title	Developing equitable, personalised community rehabilitation interventions for people with peripheral artery disease.

Research Type	Feasibility study, Qualitative study, Other (please specify)

Please provide more details on your selection above.
Participatory research: Intervention co-design

Award Duration You may undertake the award on a part-time basis between 80% whole time equivalent (WTE) and 100% WTE due to personal circumstances; this must be discussed with the NIHR before submission of the application.
60

Proposed start date if grant awarded	01/10/2025

2. Applicant CV

Degrees and Professional Qualifications

From	To	Qualification	Subject	Country	Institution	Class	Department
09/2005	09/2006	PGCHE	Postgraduate Certificate in Academic Practice	United Kingdom (England)	King's College London	Merit	Academic Department of Physiotherapy
10/1996	07/2000	PhD	The Rehabilitation of Rheumatoid Arthritis: physical, functional, and biochemical parameters	United Kingdom (England)	King's College London	Pass, with no corrections	Applied Biomedical Sciences
09/1991	07/1992	MSc	Sport Science	United Kingdom (England)	Loughborough University	Merit	School of Sport, Exercise and Health Sciences
09/1983	07/1986	Graduate Diploma	Physiotherapy	United Kingdom (England)	King's College Hospital	Distinction	Physiotherapy

Present and previous positions
(including from/to dates, job title and employer)

From	To	Position	Organisation	WTE % Research
09/2021	-	Professor of Physiotherapy & Rehabilitation	City St George's University of London	40
05/2020	-	Senior Research Fellow (Knowledge Mobilisation)	National Institute for Health and Care Research	0
09/2014	08/2021	Part time Senior Lecturer	King's College London	20
07/2004	09/2012	Part time physiotherapist	Epsom Physiotherapy and Sports Injury Clinic	0
09/2002	08/2014	Part time Lecturer	King's College London	20
07/1999	09/2002	Part time physiotherapist	St Anthony's Hospital	0
09/1996	07/2000	Research Associate/PhD student	King's College London	100
10/1995	08/1996	Band 8a Physiotherapist	Fleet Hospital, Hampshire	0
09/1992	09/1995	Band 7 Physiotherapist	Nuffield Orthopaedic Centre, Oxford	0
09/1988	08/1991	Band 6 Physiotherapist	Greenwich District Hospital	0
08/1986	08/1988	Physiotherapist	St Thomas' Hospital, London	0

Research Grants held

Title of Grant	Role in the Research Grant	Total Grant £	Source of Grant (Organisation Name)	Duration (In Months)	Start Date of Funding	End Date of Funding
Walking exercise behaviour-change in people with intermittent claudication: A	Chief Investigator. Overall responsibility for research design and delivery with a focus on key success criteria, deliverables,	2898170	NIHR PGfAR	63	01/09/2025	30/11/2030

Title of Grant	Role in the Research Grant	Total Grant £	Source of Grant (Organisation Name)	Duration (In Months)	Start Date of Funding	End Date of Funding
multi methods research programme including the	outputs & dissemination. Line management of programme staff, doctoral student (funding outcome due December 2024)					
Co-designing recommendations for community physiotherapy rehabilitation after day-case lower limb joint replacement	Co-investigator. Responsible for development of research question, design and coordination of the work packages	102331	Physiotherapy Charitable Trust	18	01/03/2025	01/09/2026
INSIGHT: Inspiring Students into Research (South London Collaboration)	Co-investigator. Responsible for St George's contribution to research capacity development	1500000	National Institute for Health and Care Research	36	01/09/2024	01/08/2027
Personalised management of patients with intermittent claudication using biomarkers and a monocyte-based therapy	Co-investigator. Responsible for design and delivery of exercise and physical activity intervention	1095678	British Heart Foundation	60	01/09/2024	31/08/2030
Implementation of Comprehensive Geriatric Assessment based perioperative medicine services to improve clinical outcomes	Co-investigator. Responsible for design and delivery of for public involvement and stakeholder engagement strategy	1439867	NIHR HS&DR	39	01/04/2024	01/07/2027
NIHR Doctoral Fellowship (Mr Gregory Booth)	Co-supervisor. Responsible for academic and clinical mentorship to Mr Gregory Booth (Clinical academic physiotherapist)	326661	NIHR Academy	36	01/01/2024	31/12/2026
National Incubator Hub for Community Rehabilitation Research	Chief Investigator and lead applicant: Responsible for development of delivery of hub engagement activites. Line managemnt of hub staff	150000	National Institute of Health and Care Research	36	01/01/2024	31/12/2026
Developing a role for patients and the public in the implementation of health and social care research evidence into practice	Co-Investigator. Responsible for knowledge mobilisation and dissemination strategy	838773	NIhr HS &DR	36	01/05/2022	30/04/2025
Health Education England Integrated	Co-investigator. Lead for Physiotherapy	150000	NIHR	36	01/09/2022	31/08/2025

Title of Grant	Role in the Research Grant	Total Grant £	Source of Grant (Organisation Name)	Duration (In Months)	Start Date of Funding	End Date of Funding
Clinical Academic (ICA) Awards						
Post-operative rehabilitation programme to improve walking in older people with neurogenic claudication	Primary Supervisor. Responsible for academic and clinical mentorship for Mrs Suzanne McIlroy (Clinical academic physiotherapist) during her Doctoral Research Fellowship	300372	Dunhill Medical Trust	48	01/10/2021	01/03/2025
Effect of −Lactoglobulin supplementation as a preoperative therapy in older people with OA undergoing hip replacement	Co-investigator. Physiotherapy clinical lead (declined)	114000	Arla International	24	01/09/2020	31/08/2022
Gait interventions for patients newly diagnosed with rheumatoid arthritis of the foot or ankle	Co-investigator: Responsible for intervention design and lead for process evaluation	1836461	NIHR HTA	60	01/09/2017	31/08/2024
Disseminating evidence to patients, practitioners and decision makers	Chief Investigator. Responsible for study design, research delivery including staff management and reporting.	107403	NIHR	24	01/09/2022	31/08/2024
Understanding how members of the public from underserved communities' access and use scientific evidence	Chief Investigator	52136	NIHR	12	01/09/2021	31/08/2022
The feasibility and applicability of remote rehabilitation provision for COVID-19 Survivors	Co-investigator: Responsible for public involvement and engagement strategy design and delivery	14231	King's College Hospital Charity	12	01/09/2020	31/08/2021
A brief physiotherapist-led behaviour-change intervention to facilitate walking in people with peripheral arterial disease	Chief Investigator. Responsible for trial design, research delivery including staff management and reporting.	299495	Dunhill Medical Trust	36	01/09/2017	30/03/2020
International Doctoral Studentship (Ms Laura Meade)	Co-supervisor for Ms Laura Meade (health psychologist) during her doctoral fellowship	116300	King's College London	36	01/10/2016	01/05/2020

Title of Grant	Role in the Research Grant	Total Grant £	Source of Grant (Organisation Name)	Duration (In Months)	Start Date of Funding	End Date of Funding
CAT Masters in clinical research programme	Co-investigator. Physiotherapy Lead	2654532	NIHR	36	01/10/2013	30/09/2016
Exploring the experience of walking in people with intermittent claudication	Chief Investigator. Responsible for study design, research delivery and reporting	11255	Guy's and St Thomas Charity	12	01/10/2015	30/09/2016
Doctoral Research Fellowship (Ms Melissa Galea Holmes)	Primary Supervisor for Ms Melissa Galea Holmes during Doctoral Fellowship	100314	Dunhill Medical Trust	48	01/10/2010	30/09/2014
The impact of fatigue and physical inactivity in people with Antiphospholipid Syndrome	Chief Investigator. Responsible for study design, research delivery and reporting	1000	APS Support UK	12	01/10/2014	30/09/2015
Education and eXercise Training in early Rheumatoid Arthritis study – (EXTRA trial	Chief Investigator. Responsible for study design, research delivery and reporting	84483	Physiotherapy Research Foundation	36	01/10/2008	30/09/2011
Upper limb sensorimotor function in patients with Rheumatoid Arthritis	Research studentship. Responsible for primary supervsion	1000	King's College London Summer Studentship	3	01/06/2024	31/08/2024

Relevant Prizes, Awards and other Academic Distinctions

Please note the information for additional prizes or awards is automatically populated from information provided in the "Update CV" section of the "Manage My Details" page, which you can update by following this link https://aramis.nihr.ac.uk/MyAccount/UpdateCV.aspx

Date (mm/yyyy)	Source	Prize	Role of Applicant
01/06/2019	European Alliance of Associations for Rheumatology	Annual research excellence award: 'The prevalence and impact of fatigue in adults with Antiphospholipid Syndrome: A mixed methods study'	Lead applicant/awardee
01/05/2015	British Society of Rheumatology	Jewels in the Crown annual research excellence award: 'Economic Evaluation of a Brief Education, Self-Management and Upper Limb Exercise Training Programme in People with Rheumatoid Arthritis'	Lead applicant/awardee
01/09/2013	Higher Education Academy	Fellowship of Higher Education Academy	Awardee
01/09/2003	Chartered Society of	Robert Williams Research	Lead applicant/awardee

Relevant Prizes, Awards and other Academic Distinctions

Please note the information for additional prizes or awards is automatically populated from information provided in the "Update CV" section of the "Manage My Details" page, which you can update by following this link https://aramis.nihr.ac.uk/MyAccount/UpdateCV.aspx

Date (mm/yyyy)	Source	Prize	Role of Applicant
	Physiotherapy	Medal for research excellence 'Improvements in motor and functional performance are maintained following rehabilitation in patients with rheumatoid arthritis'	
01/09/2002	British Health Professional in Rheumatology Clinical Prize	Award for best clinical innovation: Implementation of pre-assessment clinics for patients undergoing Surface Hip replacements.	Lead applicant/awardee
01/09/1996	Arthritis Research Campaign	Scholarship bursary to fund PhD fees	Lead applicant/awardee
01/07/1986	Chartered Society of Physiotherapy	Membership	Lead applicant/awardee

ORCID iD 0000-0002-2190-8590

3. Applicant Research Background

Professional Background	Allied Health Professional

Profession (Allied Health Professional)	Physiotherapist

Please describe your research career to date

I have an ***outstanding track record of high-quality interdisciplinary research*** in the rehabilitation of painful long-term conditions (e.g., peripheral arterial disease, PAD) that impacts beneficially on practice and patient function, activities and quality of life. As Chief Investigator or co-investigator, my grant income totals £11.8million (£8.8million since 2021, £2.9million pending).

My research is at the forefront of the national agenda to shift care into the community. I led the team that tested one of the first, home-based walking exercise behaviour change intervention for PAD patients (MOSAIC1)[1]. This research was published in JAMA and showcased in a JAMA editorial and multiple media organisations. It underpinned the inclusion of home-based exercise in the recent European PAD consensus statement[2]. MOSAIC1's feasibility and implementability are now being investigated in Scotland[3] and informed my NIHR PGfAR application (Chief Investigator, stage 2, pending).

I led the first research programme to identify poor sensorimotor function, functional performance, and quality of life in adults with rheumatoid arthritis (RA). I provided key evidence on the safety and effectiveness of exercise for reversing these debilitating effects and improving patient quality of life. My research impact was recognised by the British Society of Rheumatology (BSR Prize 2002) and Chartered Society of Physiotherapy (Silver Medal 2003). I designed clinically applicable, novel interventions and led trials determining RA upper limb exercise programme and early RA gait rehabilitation effectiveness and cost-effectiveness[4-7]. My research informed guidelines and drove the integration of exercise into RA management[8,9]. My research's excellence was recognised by the following: a BSR 'Jewels in Crown' keynote (2015), the European Alliance of Associations for Rheumatology (EULAR) research excellence award (2019) and co-authorship of the 'Oxford Textbook of Rheumatoid Arthritis (2022).

My ***collaborations*** with European experts from over 44 countries enabled participation in some of the most impactful European guideline development groups[10]. I am lead author on international outputs driving practice change[11-13]. These partnerships sparked collaborations [NIHR150711] and invitations to supervise an international PhD student[14,15]. I lead partnerships promoting community rehabilitation research. I secured funding for a consortium of over 32 researchers, professional and voluntary organisations to deliver the NIHR-supported Community Rehabilitation Research Incubator hub. This enabled participation in the Community Rehabilitation Alliance, a coalition of over 60 charities and professional bodies, driving the pace of change in commissioning, planning and delivery of evidence-based community rehabilitation nationally. We developed best practice standards and defined service level outcome datasets. I work in partnership with community and acute clinical service planners as part of our Integrated Care Board (ICB) research working group to develop research that addresses local patient needs.

My research leadership is defined by my role as an academic leader in physiotherapy and rehabilitation research.

- I am a panel member for Pre-Clinical Academic and practitioner Fellowships, Wellcome Trust Global Health Doctoral Fellowships, and former RfPB panel member, acting as lead assessor.
- I provide strategic oversight of substantial NIHR awards as D-MAPP Programme Committee Chair [NIHR201612] and Trial Steering Committee member (three trials).
- I regularly provide expert review for international rehabilitation funding calls e.g. Dutch Physiotherapy Society.
- I was the first physiotherapist Associate Editor for Rheumatology Advances in Practice and contributed to all aspects of the editorial process including editing the inaugural special issue on 'Exercise as Medicine: Physical Activity in Inflammatory Arthritis'.
- As Trustee and Chair of the British Health Professions in Rheumatology scientific committee and BSR

conference committee member, I led the research prioritisation, reviewed proposals, curated and showcased high-quality research at the annual conference.

I lead flagship ***translational research capacity building*** activities. As NIHR Knowledge Mobilisation Fellow, my team develop and deliver the national NIHR strategy for maximising NIHR research impact. I revised the application form/guidance, developed researcher knowledge mobilisation capability through national webinars and designed and delivered 'maximising impact' training for NIHR RSS advisors. I will lead national translational researcher capability building as a founding member of the NIHR-supported Knowledge Mobilisation Incubator.

I have a substantial ***track record in developing multidisciplinary researchers***. I lead the NIHR-supported Community Rehabilitation Research Incubator hub aimed at building research capability in multi-professional community practitioners. As physiotherapy lead for the HEE Integrated Clinical Academic and NIHR Insights South-East London programme, I inspire the next generation of allied health professional (AHP) researchers.

I effectively coach research students through fellowship applications, researcher development, PhD examinations and postdoctoral awards. I've supervised six PhD students to completion on time. I currently supervise three PhD students and three clinical NIHR predoctoral awardees.

Through BSR and NIHR mentoring programmes, I mentor multidisciplinary researchers at all levels including a physiotherapist to secure her first externally funded postdoctoral award and a postdoctoral health psychologist to obtain her first tenured academic position and develop her NIHR advanced fellowship application.

I am passionate about ***inclusive patient involvement*** using innovative strategies to maximise involvement from underserved communities. I established the City St George's vascular public advisory group. By partnering with community groups (e.g., national PAD advisory group, Eastern European and Romanian national community group), we co-developed creative engagement approaches. Our group now involves geographically diverse patients from PAD underserved communities (e.g., Hull, Leicester, London). Patients drive the identification of research questions and partner us to deliver impactful research. For example, with my patient advisors, my current research has shifted to address the needs of underrepresented PAD groups such as people with chronic limb threatening ischaemia and people from socioeconomically deprived or some ethnic communities. To enhance involvement, I provide technology support, plain and multiple language summaries, research training and reimburse expenses.

This NIHR Professorship will enable a step-change into leading multiple research projects of international importance and a strategic leadership role aligned with the NIHR priorities[16]. This professorship will provide me with the support I need to develop my long-term vision: to radically shift the scientific understanding, quality of care and equity of access to PAD community interventions. The financial and institutional commitment offered by the scheme and City St George's will enable me to further expand my research group, increase research activity, support community rehabilitation research capacity building and maximise impact.

Has this application, or a similar application, previously been submitted to this or any other funding body?

Yes

If Yes, provide details

Workstream 3 of this NIHR Professorship application was submitted as the first three workstreams of an NIHR Programme Grant for Applied Research. The application is under consideration at Stage 2 and the outcome is expected in December 2024. Funding for workstream 3 activity has been included in both applications.

Contextual Factors

Please use this question to detail any contextual factors you wish to make the Selection Committee aware of. NIHR wants to know about any circumstances so that they may take them into consideration during the assessment of your application. contextual factors may include:

- Career breaks due to parental leave, or periods of illness.
- Reduced time spent undertaking research due to a disability or caring responsibilities. This could include any physical or mental difficulty that may have impacted your research career. These are situations that have a significant impact on your ability to undertake research.
- Reduced opportunities to career support e.g. mentorship, and limited opportunities to undertake prior research and training.
- Impact of the COVID-19 pandemic on your research career.

Please also use this section to detail any other factors that may have impacted your research career not listed in

the examples provided. The impact on your career to date will be specific to your particular circumstances but could include such impacts as limited opportunities to obtain grant funding, or fewer publications. In general terms, mitigating factors should be significant, and relevant.

NIHR acknowledges that you may be reluctant, or uncomfortable disclosing relevant information that is sensitive. However, you should bear in mind that we are unable to take into account factors that you do not disclose. Please be assured that information provided by you is sensitive and will be treated confidentially and in line with General Data and Protection Regulations (GDPR).

Commencement of research career. My research career builds on my substantial experience as a clinical physiotherapist. I was inspired to develop my research career when I was unable to identify high quality evidence to underpin my practice. I was determined to improve the physiotherapy evidence base, despite it not being the prevailing culture or environment. As rehabilitation research was in its infancy and physiotherapy research roles/mentorships were limited, my research career was largely self-driven. I self-funded an MSc Sports Science at Loughborough University and, after returning to full time clinical practice for 5 years (with no dedicated research time), I obtained a small Arthritis Research Campaign bursary (£6,000) to fund PhD fees.

Postdoctoral research opportunities. I completed my PhD viva when pregnant with my first child. I returned to clinical practice after parental leave, and did not complete any dedicated post-doctoral research. I commenced an educational position in a university teaching department with minimal protected research time.

Maternity leave and caring responsibilities. I have had two periods of maternity leave (one year each). One immediately after I completed my PhD and another after completing my first year as a (teaching) lecturer.

Part-time working and educational/administration load. Due to caring responsibilities, I worked as a part-time teaching lecturer (0.5 WTE,2002-2013), then Senior Lecturer (0.6 WTE 2013-2021). The primary focus of my department was delivering three teaching intensive AHP programmes. Educational activities accounted for at least 60% of my part-time roles. I delivered a substantial teaching load, led three teaching modules, chaired the postgraduate physiotherapy programme board and was postgraduate research co-ordinator/chair of the postgraduate committee. For one year, I was Programme Director for the MSc in Advanced Physiotherapy (0.8 WTE, 80% education). Whilst there were limited opportunities for research collaboration; to write research funding applications or publications, these positions provided a stepping stone into more research intensive roles and fuelled my ambition to develop the next generation of clinical academic AHPs.

To increase research time and support, I changed institutions in 2021. City St George's provided more research time, development opportunities, leadership training and encouraged me to apply for this NIHR Professorship. I have increased outputs, postgraduate students, research networks and funding whilst also being seconded as a part-time NIHR Senior Research Fellow (Knowledge Mobilisation). My NIHR secondment has developed my strategic planning and leadership skills and widened my national and international networks, but has impacted on my research time, which remains restricted.

Together, these circumstances provided a secure foundation for a career in applied research. However, they impacted on my research trajectory including number of outputs, PhD students, collaborations and the delivery of my ambitious research programme. This award will transform my career. It will release me from teaching and my NIHR secondment and will enable me to accelerate my team building. The beneficiaries will be patients, clinicians, the NHS and the academic community. This application has a secure foundation, with guaranteed deliverables, alongside explorative transformative projects.

4. Plain English Summary of Research

Why is this research needed?

Out-of-hospital rehabilitation (known as community rehabilitation) improves people's health and wellbeing and saves money. But community rehabilitation is not provided equally, and it is not known how best to treat people with some conditions or whether treatments are acceptable to patients. Improving equal access to high quality community rehabilitation is central to the government's NHS plan.

One of the conditions that we don't know how to rehabilitate in the community is called peripheral artery disease. Peripheral artery disease is poor leg blood flow due to narrowed arteries. People from Black, African Caribbean, South Asian or Eastern European communities, those with diabetes or living in areas with socioeconomic deprivation are affected more by peripheral artery disease.

Severe peripheral artery disease can cause constant leg pain, sores and gangrene in the toes. This is called chronic limb threatening ischaemia. Most people need their arteries unblocked in hospital and then need to self-manage their recovery at home. But there are no self-management programmes to support people with chronic limb threatening ischaemia to recover and live well at home.

Some people with milder peripheral arterial disease experience leg pain on walking (called intermittent claudication). Walking frequently is the best way to improve this pain and reduce the risk of severe artery disease. It is hard for people to walk on their own when they are in pain. New ways are needed to help people walk regularly, improve their leg pain and do more activities.

Many people with peripheral artery disease are not active enough to reap the health benefits. This increases their risk of getting other health problems. There are many ways to help people to become active. One way is to use a smartwatch that records how much activity is done and sends messages to encourage people to be more active. Smartwatches can be used separately or as part of an activity programme. But some older people find smartwatches difficult to understand and use.

Aims

To work with patients, practitioners and service planners to co-design ways to provide community rehabilitation equitably for everyone with peripheral artery disease and explore new ways to help people to be more active. People from groups who are most affected by peripheral artery disease and underrepresented communities will be included.

What will happen?

I will build a team to

1. summarise the evidence for the factors that influence self-management, interview patients and practitioners and co-design a self-management programme to help people with chronic limb threatening ischaemia live well at home.
2. look at the results from a study of an existing walking exercise programme for people with intermittent claudication and use this information to adapt this programme so it can be provided in the community.
3. interview patients and practitioners, co-design and user test a new treatment including a smartwatch to help everyone with peripheral artery disease be more active.
4. co-design a plan to implement these treatments in the NHS, if these ways are shown to be effective and acceptable.

How have patients and communities been involved?

Members of the public and patients from different backgrounds chose this topic and helped design this research. They helped write this summary. The public advisory group will meet every 6-months and be involved in co-design and workstream management.

Sharing this research

Patients, practitioners, service planners and voluntary and professionals' organisations (like Circulation

Foundation) will co-design and help share our research progress and findings, so it is available in the NHS and for the public. This may include writing research summaries; giving talks; hosting meetings; designing infographics, animations, websites and social media.

5. Scientific Abstract

Provision of community rehabilitation is inequitable. Many interventions are not evidence-based or acceptable to patients or providers. Rehabilitation of some high burden conditions, such as peripheral artery disease (PAD), receive relatively little national support, prioritisation or research funding, fuelling health inequalities.

PAD affects over 237 million adults globally. One common PAD symptom is intermittent claudication (IC) although in its most severe manifestation, it leads to chronic limb threatening ischaemia (CLTI). Over half of those with symptomatic PAD will die, have an amputation or cardiovascular event within five years of diagnosis. CTLI prognosis is extremely poor; commensurate with terminal cancer. The economic burden of PAD is substantial and rising.

PAD patients report dissatisfaction with care and improving outcomes across the PAD symptom range is the highest ranking James Lind Alliance research priority among all vascular conditions. Providing PAD rehabilitation and supporting self-management will save money. Yet, equitable access to community rehabilitation is compromised by lack of acceptable, feasible interventions that were co-designed with patients and providers, restricted resources and a high and increasing demand from a diverse patient group with complex needs. Many people with PAD are also excluded from research and unable to access or use new treatments e.g., digital interventions.

New scalable, co-designed, inclusive PAD interventions such as provider supported community rehabilitation, self-management programmes and 'next-generation' mobile-based health technologies (mhealth) are promising. Yet, evidence is mixed, none have been designed with PAD patients and providers, or commissioned in the NHS.

The aim of this professorship is to transform the care for people with peripheral artery disease through community rehabilitation interventions. Over five work packages, I will lead research that will:

1. Maximise diverse and inclusive involvement, participation and engagement of PAD underserved groups.

2a. Through systematic review, qualitative and participatory approaches, codesign an inclusive supported self-management intervention to improve quality of life in CLTI patients.

2b. Develop trial protocol, funding application, set-up/commence a trial.

3a. Through secondary analysis of existing quantitative and qualitative data, participatory approaches, co-refine and feasibility test an inclusive home-based walking exercise behaviour-change programme for people with IC in primary and community services.

3b. Develop trial protocol, funding application, set-up/commence a trial.

4a. Through qualitative approaches, co-design, adaptation and user testing, develop an inclusive mhealth intervention for PAD patients to increase physical activity.

4b. Develop trial protocol and submit funding application.

5. Drawing on data from workstreams 2-4, specify the implementation context, outcomes and mechanisms and co-design a programme theory, preliminary implementation and evaluation strategies for inclusive PAD community rehabilitation services.

This ambitious programme of work balances building on my existing work with guaranteed high impact outputs, and projects which promise a rapid upward trajectory within five years. It will spearhead the shift of PAD care into the community and develop research capacity in under represented disciplines.

Mitigating research and digital exclusion and involving patients and the public are central to this research. Patients identified the research questions, helped plan this research and will help deliver and mobilise research progress and findings using co-designed, creative approaches.

6. Detailed Research Plan

Please complete this section paying attention to the ten suggested subheadings/sections outlined in the guidance document.

The problem

Globally, one in three people require rehabilitation and ageing populations and increasing numbers of people living with multiple long-term conditions are fuelling demand[17]. In 2017, the World Health Organisation recognised that many countries, including the UK, are not providing appropriate and equitable rehabilitation nor are they equipped to address future rehabilitation needs. Their call to action urged countries to scale-up evidence-based rehabilitation to address health and social inequities as a critical health priority[17].

Community rehabilitation is person-centred, out-of-hospital, care tailored for people with different conditions and fluctuating needs. UK provision of community rehabilitation is inconsistent with many interventions not evidence-based, feasible or acceptable to patients or providers[18]. Rehabilitation of some high burden conditions, e.g., peripheral artery disease (PAD), have received relatively little national support, prioritisation or research funding compared to other cardiovascular diseases (e.g., coronary heart disease), fuelling health inequalities[19].

Peripheral artery disease affects over 237 million adults globally and 15-20% of the UK population aged ≥55 years[20,21]. It frequently occurs alongside other conditions e.g., diabetes, high blood pressure or cholesterol[22,23] and disproportionally affects socioeconomically deprived populations, and South Asian, Black, African Caribbean, and Eastern European groups[22,24-26]. It causes substantial disability, increased cardiovascular morbidity and mortality and is associated with feelings of powerlessness, isolation and poor quality of life[21,23,27,28]. Its economic burden is substantial; cardiovascular-related healthcare costs alone in England for PAD are £1.4 billion/year, with wider economic annual costs approximately £4 billion[29,30].

One PAD symptom is walking leg pain (intermittent claudication, IC) although in its most severe manifestation, PAD leads to rest pain, and tissue loss (chronic limb threatening ischaemia (CLTI))[31]. Despite optimal medical treatment, over half of those with symptomatic PAD will die, have an amputation or cardiovascular event within five years of diagnosis[22,23,27,32,33]. CTLI prognosis is particularly poor, even after revascularisation; commensurate with terminal cancer[34].

Improving PAD outcomes is the highest ranking James Lind Alliance research priority among all vascular conditions[35].

PAD care is delivered through a network of secondary/tertiary hub hospitals supported by community/primary care spoke services. Guidelines recommend medical optimisation, lifestyle modification, revascularisation/limb salvage, community rehabilitation and home-based self-management support (including personalised home-based exercise, digital interventions)[36,37,38]. Yet, equitable access to community rehabilitation that develops an individual's capability, opportunity and motivation to self-manage their symptoms and condition at home is compromised by lack of acceptable, evidence-based interventions and a high and increasing demand from a diverse patient group with complex needs[18,39]. Many interventions are not designed with patients or providers and are impracticable for providers to deliver due to restricted knowledge, skills or resources (technology, time, space, training, staff)[18].

PAD community rehabilitation is an area of historical under investment and high research need.

Equity factors influence access to healthcare, health and research opportunities and outcomes. About 24% PAD patients screened for eligibility are excluded from participating in rehabilitation trials due to at least one equity factor[40,41]. Yet, these subgroups may respond differently to community rehabilitation due to differences in social, environmental, physiological or disease states. This is important because findings from PAD trials and systematic reviews inform practice guidelines and service planning so excluding these subgroups may mean that community interventions may not be as effective once implemented.

These gaps need urgently addressing. Subgroups with higher prevalence or disease burden, or those not receiving adequate services need including in PAD intervention development and research so evidence-based community rehabilitation interventions can be implemented equitability and rapidly at scale.

Importance

PAD patients report dissatisfaction with care[42]. Low engagement (50% enrolment) and high attrition (30% non-completion) in hospital-based PAD rehabilitation is driven by social (e.g., family support), physical (e.g., pain), and environmental factors (e.g., travel)[43,44,45]. Women, older adults, CLTI patients, people from socioeconomically deprived or minority ethnic communities are disproportionately affected by these barriers[46,47]. New scalable, co-designed, inclusive interventions are needed such as practitioner supported community rehabilitation, self-management programmes and mobile-based health technologies (mhealth)[48,49]. These approaches will save money as patients most confident in managing their long-term conditions have 38% fewer emergency admissions and 32% fewer emergency hospital attendances than those who feel least confident[50].

Public advisors and practitioners support this research because it addresses avoidable PAD and health deterioration from inequitable NHS access. It focusses on five James Lind Alliance PAD priorities; How can we a) improve patient outcomes b) make it easier for patients to get help for this problem? c) help better educate patients with poor leg circulation? d) improve exercise provision and access? and e) what are the best ways to reduce leg pain without an operation?[35] It aligns with National Institute for Health and Care Excellence (NICE) [CG147] research recommendations (e.g., optimal exercise therapy for PAD)[36] and NIHR strategic themes[16].

Crucially, it is congruent with the Department of Health and Social Care (DHSC) areas of research interests (early action to prevent poor health outcomes, accelerating innovation adoption and scale-up)[51], NHS digital health strategy[52] and recent Darzi recommendations to: shift care to community; expand digital services; promote prevention and minimise disease progression[53].

Existing evidence

Equitable PAD rehabilitation requires a suite of evidence-based inclusive community interventions considering the severity, complexity and biopsychosocial impact of PAD and associated multiple comorbidities on everyday life.

People with CLTI have severely restricted lifespan, high symptom severity and co-morbidities resulting in poor quality of life and high social isolation, even after revascularisation[27]. Systematic reviews support the effectiveness of short self-management approaches in people with long-term and life-limiting conditions on quality of life and independence[54,55,56]. When people receive personalised support, their emotional wellbeing improves because they feel more in control and gain more knowledge and strategies to manage everyday life[54-56]. Evidence for self-management interventions emphasised the need for contextualisation to 1) specific CLTI challenges and complexity and personal and cultural characteristics 2) understanding the community setting and 3) training addressing providers' knowledge, skills and attitudes[57,58]. No community supported self-management interventions have been co-designed with CLTI patients and providers or commissioned in the NHS.

Guidelines recommend supervised exercise programmes as first-line treatment for IC as evidence is insufficient to recommend home-based walking exercise, given inconsistent trial results[36,59]. Systematic reviews concluded that home-based interventions are inferior to hospital based exercise but better than providing no or basic exercise advice[60,61]. Most studies were small and of low or very low quality. Heterogeneous intervention delivery methods,

settings, intervention components not including tailored behaviour change techniques, exercise dose variability, impractical interventions, inappropriate outcomes and/or poor treatment fidelity all contribute to uncertainty about effectiveness of home-based exercise interventions.

MOSAIC1, a feasible, acceptable home-based PAD walking exercise behaviour-change intervention, addresses these limitations[1,62-64]. In an efficacy trial, participants receiving MOSAIC1 increased mean 6-minute walk distance by a clinically meaningful difference, compared with usual care at 3-months[1]. MOSAIC1 needs adapting for delivery by community providers to diverse patients across the UK and long-term effectiveness and cost-effectiveness established. This informs my PGfAR application (Stage 2-pending).

Physical activity is a tertiary prevention in PAD reducing the need for revascularisation, and post-revascularisation complications in those requiring surgery, as well as the need for further complex, costly interventions. Yet, less than 50% of PAD patients meet physical activity guidelines for general cardiovascular health or for improving IC[65,66,67].

The COVID-19 pandemic prompted community services to incorporate mhealth to deliver care and offer preventative strategies such as promoting physical activity. Yet, older people, including those with PAD, often experience digital exclusion[49]. Whilst mhealth interventions increase physical activity in the general population, the effects in PAD patients are unclear[68]. A Cochrane review comparing mhealth interventions versus usual care, exercise advice or supervised exercise included only four trials and found no evidence of mhealth interventions' effectiveness in people with IC[48]. All studies had unclear or high risk of bias, excluded CLTI patients or people post-revascularisation, and two enrolled majority white participants, limiting generalisability[48].

Mhealth interventions now combine consumer activity trackers (i.e. Smartwatches) with 'just in time adaptive interventions' (JITAI) – personalised motivational messages delivered in response to activity feedback[69,70]. Trials incorporating mhealth interventions that include strategies to mitigate digital exclusion show promising results[71]. Whilst a PAD message library has been developed, these are untested[72]. These 'next generation' mhealth interventions potentially provide an exciting, low-cost, adaptable and scalable PAD treatment. If co-designed with PAD patients and providers and integrated with digital inclusion strategies, they could offer either a standalone community intervention or be incorporated into interventions like MOSAIC1 to support long-term self-management.

Why is this research needed now?

My research has influenced practice and now needs accelerating to address key underserved groups - those with CLTI and those experiencing research and digital exclusion. It addresses a crucial unmet health need, the avoidable deterioration in PAD and health from inequitable access to community rehabilitation. It will provide new community interventions and workforce training for integration into the PAD care pathway, adaptable to other contexts and populations.

<u>Aims and objectives</u>

Aim: To transform the care of all people with PAD across the symptom range through community rehabilitation interventions

Objectives:

1. Maximise diverse and inclusive involvement, participation and engagement of underserved PAD groups.

2a. Through systematic review, qualitative and participatory approaches, co-design a supported self-management intervention to improve quality of life in CLTI patients.

2b. Develop trial protocol, funding application, set-up/commence trial.

3a. Through secondary analysis of existing quantitative and qualitative data, participatory approaches, adapt and feasibility test a home-based walking exercise behaviour-change programme for people with IC in primary/community services.

3b. Develop trial protocol, funding application, set-up/commence trial.

4a. Through qualitative approaches, co-design, refine and user test an inclusive mhealth intervention for PAD patients to increase physical activity.

4b. Develop trial protocol and funding application.

5. Drawing on workstream 2-4 data, specify the implementation context, outcomes and mechanisms and co-design preliminary programme theory, implementation and evaluation strategies for inclusive community PAD rehabilitation.

Project Plan

With my public advisory group and collaborators, I developed a programme theory that proposes the context, activities, outputs outcomes and impact of this professorship (Figure 1).

This research is informed by key frameworks (e.g., Medical Research Council framework[73], INCLUDE frameworks[74,75], updated Consolidated Framework for Implementation Research (CFIR)[76], mHealth development and evaluation framework[77]).

Workstream 1 Research Inclusion

Question: What engagement strategies are needed to reach to PAD patients, particularly those from underrepresented communities?

Aim: To maximise inclusive involvement, participation and engagement of PAD underserved groups.

Inclusion in community rehabilitation research is central to this professorship, guided the EqIA[78] and INCLUDE frameworks[74,75]. This cross-cutting workstream focusses on creative engagement activities with PAD under-served communities (i.e., socioeconomically deprived, or South Asian, Black, African Caribbean, Eastern European communities) so their unique perspectives are heard and inform all aspects of this research.

Research inclusion strategies comprise:

- Appointing a dedicated research inclusion and public partnership co-ordinator. They will focus on developing trust and understanding between researchers and our underserved groups**.**
- Co-designing culturally appropriate engagement strategies, interventions, resources and study documentation with diverse communities to realise our vision for excellent and equitable opportunities.
- Inclusive recruitment strategies that promote participation from underserved groups and those not usually

recruited into research. This will be achieved by developing creative and inclusive communications with underserved communities and patient organisations to share research details and support participation.
- Rewarding and recognising involvement, participation and engagement.
- Working with geographically diverse Local Research Delivery Network (LRDN) (e.g., urban, rural, north, south, coastal) and building bespoke engagement strategies with primary care and community services that serve underserved populations.
- Reviewing each workstream's participant demographic data and discussing how our strategies are being implemented and challenges overcome with public advisors. Additional targeted efforts and strategies will be applied to increase representation, where needed.
- Offering language translation, technical support, software fees for online meetings. Offering alternative ways of engaging e.g. telephone, receiving paper-based information.
- Ensuring all research team members, recruiting staff and providers compete cultural competency and inclusive research training.
- Tailoring knowledge mobilisation plans with different communities and promoting inclusive communications e.g., infographics, videoclips shared across many platforms.
- Regular monitoring/reporting of EqIA[78] and involvement (reported using GRIPP-2[79]), participation and engagement rates from underserved groups.

Output: (i) Diverse and meaningful engagement with PAD underserved groups in all workstreams.

Workstream 2 Inclusive supported self-management for people with CLTI

Question: Can a supported self-management programme to improve quality of life be co-designed for CLTI patients?

2.1 Aim: To synthesize the determinants of health self-management in people with CLTI.

Design: Systematic review.

Methods: Databases will be systematically searched for quantitative studies reporting determinants of self-management in CLTI. Deduplication, screening, quality appraisal and data extraction (e.g., setting; population; self-management approaches; determinants of self-management) will be completed by two researchers. Determinants will be summarised using the Theoretical Domains Framework and inform an intervention programme theory[80].

2.2 Aim: To understand the experience of CLTI and self-management approaches from patient, carers/relatives and provider perspectives.

Design: Qualitative study.

Participants: A purposive sample of ≥30 patients, ≥30 carers/relatives and ≥15 multidisciplinary providers/service planners.

Patient recruitment will draw on the EqIA[78], INCLUDE frameworks[74,75] and workstream 1. Diverse patient participants (i.e., location, disease severity, protected characteristics including age, ethnicity, sex) and carers/relatives will be recruited. No upper ages limits will be applied and patients with cognitive impairment will be included (with nominated supporter) if they have capacity to consent. Potential participants will be identified via vascular centres, primary and community services, social media, community organisations (e.g., National vascular public advisory group, national Romanian and Eastern European Hub), our networks (e.g., Centre for Ethnic Health Research) and NIHR LRDN's with socioeconomic, ethnic and geographical diversity (South-east and south-west London, Northeast and North Cumbria: Humberside, East Midlands).

Providers will be identified from primary/community services, professional networks, interest groups (Community Healthcare Alliance of Research Trusts, Community Rehabilitation Research Incubator, Vascular Society of Great Britain and Ireland, Society for Academic Primary Care) and social media.

Service planners will include clinical leads, ICBs, Primary Care Network (PCN), GP Cluster and Health Board, and AHP representatives from urban and rural geographically diverse ICBs, primary/community services and professional networks.

Methods: Online/in-person/telephone narrative individual/dyad patient/carer interviews (participant preference) and online/in-person provider focus groups (n=6-8 participants/group) or interviews will explore experiences of CLTI and receiving or delivering self-management. Themes will be identified using reflexive thematic analysis and participants' accounts used to develop a film of patient experiences and priorities[81].

2.3 Aim: To co-design inclusive self-management resources for anyone with CLTI and provider training.

Design: Working with Bridges Self-management (Bridges), experienced-based co-design methods will underpin the co-design of a tailored supported self-management programme for CLTI and provider training resources. Bridges is a social enterprise led by Professor Fiona Jones that has a 15-year international research programme developing personalised self-management support in many healthcare areas[57,82].

Participants: A purposive sample of ≥15 patients, ≥15 carers/relatives and ≥15 providers/service planners (see 2.2).

Methods: Participants will meet initially in separate online/in-person patients/carer or professional groups to show the film, discuss findings from 2.1 and 2.2 and identify priorities/needs/ideas. They will meet together to agree priorities and generate the intervention resources and provider training content. They will explore potential implementation barriers/facilitators. A preliminary programme theory will be developed and materials for patients and provider training finalised. The intervention and provider training will be specified using TIDieR[83].

2***.4 Aim: To co-design and commence a trial to evaluate the CLTI self-management intervention.***

Working with public advisors and collaborators, we will co-design a trial protocol, submit an external funding application and commence a trial to investigate intervention effectiveness, cost-effectiveness and implementability.

Output: (i) Co-designed CLTI self-management intervention. (ii) trial protocol and funding application (iii) trial set-up/commencement.

Workstream 3 Supported home-based walking exercise behaviour change programme for people with intermittent claudication

This workstream builds on the MOSAIC1 trial[1]. It is costed in both this professorship and my PGfAR application (Stage 2-pending).

Question: Can a home-based walking exercise behaviour-change programme delivered in primary/community services improve walking capacity in IC patients?

3.1 Aim: To identify core determinants of long-term walking exercise in IC patients.

Design: Mediation analysis.

Methods: Mediation analysis of MOSAIC1 trial (n=190, 6 Southeast England sites) psychological process variables (Theory of Planned Behaviour, Illness Perceptions)[84,85] to identify determinants of long-term walking exercise informed by behaviour change theory[86].

3.2 Aim: To Identify factors that influence walking exercise and implementation barriers/facilitators from patient/provider perspective.

Design: Framework analysis.

Methods: Analysis of interviews from 20 MOSAIC1 participants and 14 physiotherapists to identify critical factors influencing long-term walking exercise, intervention delivery and implementation barriers/facilitators informed by behaviour change and implementation theories[1,62,76,86].

Using findings from 3.1-3.2, public advisors will help prioritise walking exercise determinants[86], implementation factors[76], refine the MOSAIC1 programme theory (figure 2) and identify core behaviour change techniques[87].

3.3 Aim: To adapt MOSAIC2 for delivery in primary/community services.

Design: Participatory co-design approach.

Participants: ≥15 patients (including MOSAIC1 participants), ≥15 carers and ≥15 primary/community provider/service planners (see 2.2, 2.3).

Methods: Using the ADAPT framework to identify the key features that require adaptation[88], participants will initially work in separate groups and then together to adapt MOSAIC1, consider potential implementation barriers/ facilitators in primary/community services by a range of healthcare professionals and other exercise providers (e.g., fitness instructors, rehabilitation technical instructors, healthcare assistants). MOSAIC2 intervention, provider training/materials will be described using TIDieR[83].

3.4 Aim: To assess the feasibility of conducting the MOSAIC2 trial in primary/community services.

The feasibility of recruitment and randomisation was tested in MOSAIC1 trial[1]. This study therefore focusses on examining fidelity, acceptability and implementation outcomes and determinants of MOSAIC2 and provider training in primary/community services.

Design: Prospective single-arm study.

Participants and setting: 32 adults aged ≥50 years with IC from 6 geographically diverse sites over 6-months (see 2.2).

Interventions: All participants will receive MOSAIC2. It is envisaged that MOSAIC2 will comprise up to 6 in-person/remote/telephone sessions (patients' preference) delivered over 3-months by trained providers.

Assessment timepoints: Baseline and 3-months (post-intervention).

Outcomes: Fidelity of delivery, intervention components and proficiency of motivational interviewing, fidelity of receipt/enactment assessed in a random sample of audio-recorded sessions. Completeness of data collection including 6-minute walk distance and physical activity level using accelerometery. Acceptability and implementability assessed using mixed methods. Inclusive recruitment practices assessed by diversity of participant characteristics.

3.5 Aim: To co-design and commence a trial to evaluate MOSAIC2.

With public advisory group and collaborators, co-design a trial protocol, external funding application and set-up/commence trial.

Outputs: (i) Adapted walking exercise community rehabilitation intervention for IC patients (ii) trial protocol, external funding application (iii) trial set-up/commencement.

Workstream 4 Co-designing an mhealth intervention to promote physical activity

This workstream will be delivered with new collaborator: Dr Tomas Vetrovsky (see collaborator section).

Question: Can an inclusive mhealth intervention be co-designed to promote physical activity for PAD patients?

4.1 Aim: To explore preferred activity, sedentary behaviour patterns, potential barriers/facilitators of using mhealth interventions in PAD patients/carers.

Design: Qualitative interviews.

Participants: A purposive sample of ≥30 patients and ≥30 carers/relatives (see 2.2).

Methods: Online/in-person/telephone semi-structured individual/dyad interviews (participant preference) to explore meaningful activity, sedentary behaviour patterns, experiences and potential barriers/facilitators of using mhealth interventions including activity trackers. Common themes will be identified using reflexive thematic analysis and key behavioural targets identified using behaviour change theories[81,86].

4.2 Aim: To explore providers' experience of mhealth interventions in the community, useful features and potential implementation barriers/facilitators.

Participants: A purposive sample of ≥15 multidisciplinary providers (see 2.2).

Methods: Two online 1.5 hour focus groups (6-8 participants each) will explore experiences of using mhealth and activity trackers in community interventions, useful/redundant features, potential implementation barriers/facilitators. Themes will be identified using reflexive thematic analysis[81].

4.3 ***Aim: To co-design an inclusive mhealth intervention, including a personalised message library.***

Design: Participatory co-design approach.

Methods: Drawing on 2.3 and 4.1-4.2, ≥15 diverse patients, ≥15 carers and ≥15 providers will work in small co-design groups and together to review key behavioural targets, intervention format, components, support and communication channels. A key focus will be on digital inclusivity and accessibility. A published library of PAD physical activity behaviour change messages will be reviewed and tailored (100-150 messages)[72]. A co-design sub-group will identify key provider training components.

4.4 Aim: To explore comprehension and perceived usefulness of messages.

Design: Cognitive 'think aloud' interviews.

Participants: ≥15 diverse PAD patients (see 2.2).

Methods: Using cognitive interviewing, participants will each read 30-40 randomly selected messages and verbally express their thoughts, interpretations and perceived usefulness of the messages[66,69]. Responses will be analysed thematically using an adapted coding category[69].

4.5 Aim: To adapt and integrate the prototype mHealth intervention into an existing digital platform.

Design: Software engineering and system integration.

Methods: The prototype intervention will be implemented using an existing mHealth platform, e.g., HealthReact, developed by collaborators at the University of Hradec Kralove, Czech Republic. This platform will continuously collect and evaluate data from a linked validated consumer activity tracker (e.g., Garmin) and trigger tailored messages. Messages can be triggered at specific times and places; in response to patient's behaviour (e.g., after prolonged sitting or commencement of walking) or daily/weekly activity level. The platform allows configuration of triggering parameters, e.g. duration between messages, number of messages/day, monitor and automatically syncs with a cloud service accessible via a public Application Programming Interface.

4.6 Aim: To refine prototype mhealth intervention.

Design: Mhealth intervention refinement using plan-do-study-act cycles.

Methods: ≤15 PAD patients will use the prototype intervention for one week. Quantitative (e.g., How much was the intervention used? activity after messages) and qualitative (e.g., Ease of use? messages timing and usefulness?) feedback will be collected. Iterative refinement cycles will continue until a finalised prototype is achieved.

4.7 Aim: To explore the acceptability, inclusivity and immediate effect of finalised mhealth intervention.

Design: prospective single arm study with embedded micro-randomisation and ecological momentary assessment (real-time data collection) sub-study.

Participants: ≤30 adults with PAD and 12 providers over 4-months from 6 community sites (see 2.2).

Intervention: All participants will receive the 6-week mhealth intervention by a trained provider to support inclusivity.

Outcomes: At baseline and 6-weeks including objective physical activity measures, Walking Impairment Questionnaire[89], Physical Activity Enjoyment Scale[90], Walking Self-efficacy Scale[91,92], quality of life[93], symptom severity (e.g., Claudication Pain Scale[94]) and acceptability (Acceptability of Intervention Measure, the Intervention Appropriateness Measure, and Feasibility of Intervention[95]). Interviews with 50% participants and providers exploring inclusivity, acceptability and implementability.

Sub-study: Micro-randomisation will randomly trigger 50% messages. Surveys will explore immediate views on message relevance/usefulness. Short-term impact of messages on activity and perceived message relevance/ usefulness following triggered versus non-triggered messages will be collected.

4.8 Aim: To develop trial protocol, and funding application to investigate the mhealth intervention.

With public advisors and collaborators, a trial protocol and funding application to investigate inclusivity, effectiveness and cost-effectiveness of the mhealth intervention alone or integrated into an existing intervention (e.g., MOSAIC2) will be co-designed.

Outputs: (i) prototype PAD mhealth intervention (ii) trial protocol and external funding application.

Workstream 5 Development of a preliminary programme theory, implementation and evaluation strategies

This workstream will be delivered in collaboration with Professor Fiona Jones (mentor).

Question: How can our understanding of the context, mechanisms and outcomes be integrated into a programme theory, implementation and evaluation strategies to support spread and sustainability of PAD community interventions?

Aim: To co-design preliminary programme theory, implementation and evaluation strategies for inclusive community-based PAD rehabilitation.

Design: Participatory co-design approach.

Participants: (i) ≥15 PAD patients and ≥15 carers, (ii) ≥30 local/national community providers/service planners.

Methods: Drawing on workstream 2-4 data, key implementation determinants for successful and sustainable delivery of each intervention will be summarised using the CFIR[76].

Two workshops will be conducted. In the first workshop, patient/professional groups will (i) review and discuss the implementation barriers/facilitators for each intervention (ii) prioritise the barriers/facilitators considering their importance and feasibility in different community settings. In the second workshop, all attendees will consider implementation and evaluation strategies, generated by the ERIC-CFIR Implementation matching tool[96], and brainstorm additional strategies to address prioritised barriers/facilitators. Strategies will consider operationalisation, accessibility and inclusivity. A programme theory describing route to impact will be developed.

After the meetings, the preliminary programme theory, implementation and evaluation strategies will be collated[97] and circulated to participants for feedback.

Outputs: (i) preliminary programme theory (ii) implementation and evaluation strategies for PAD community rehabilitation interventions.

Timeline: See Gantt Chart.

Outputs, dissemination, impact

Outputs

Policy and service recommendations

If effective, this research will transform PAD community rehabilitation by providing interventions, provider training, programme theory and implementation and evaluation strategies for service planners, and local and national policy

makers. Progress and findings will be succinctly summarised in reports, presentations, NIHR alerts and case studies.

Practitioners, professional societies, and translation to NHS/international care

This research will produce scalable community rehabilitation for adaptation for different contexts and populations. Free provider training and intervention resources will be shared online and presented to community and vascular providers via training sessions and conferences regionally and nationally. Through international collaborations, societies and networking e.g., World Rehabilitation Alliance, we will contextualise and raise our research visibility through presentations and in-person meetings to inform care internationally.

Public, voluntary organisations and communities

Our research will be mobilised with participants, patient/public organisations via webinars, summaries, infographics, and animations in multiple languages. These will be accessible on YouTube and social media. My team will host community meetings and share information via faith centres, community/voluntary organisations (e.g., national vascular public advisory group) as these are important channels to underserved communities.

Academic outputs

Open-access publications and conference presentations/abstracts will target national/international vascular, primary/community services, public health, health services audiences. A study website and online repositories will collate outputs and publicise progress. We will build research capacity in an underrepresented disciplines.

Engaging with patients, the NHS, and wider population

Stakeholder mapping will identify the people and communities with the highest influence and interest in how this research progresses and findings are interpreted and implemented. With public advisors, providers and research team, a tailored mobilisation and communication strategy for each stakeholder group will be developed. Knowledge mobilisation approaches informed by appropriate theories and frameworks will be used[98]. This will include updating participants, public and patient organisations using creative methods such as blogs, webinars, infographics and social media (e.g., PAD Facebook group). I will work directly with Health Innovation Networks (HIN), Applied Research Collaborations (ARC), and continuously update service planners at ICBs and NHS England using briefings, via RD forum conference and existing meetings.

Future funding and support

Funding for at least two trials will be sought during this professorship. A decision on a stage 2 PGfAR application that includes workstream 3 and a trial is pending. As care moves from secondary to community care, service planners will be focussed on scalable interventions, costs and outcomes[53]. I will minimise costs, resulting in long-term healthcare system savings. Free intervention materials and training will be deliverable at scale. No income generation is planned.

Possible barriers for further research, development, and implementation

I will lead studies which address crucial questions, meaning the findings are more likely to be implemented. Potential barriers to research and digital participation will be mitigated by a dedicated workstream (workstream 1) and by collaborating with experts with a track record of developing inclusive mhealth interventions If effective, new NHS community services and workforce training would require commissioning; this will need resource for rollout.

Impact of this research and for whom?

This research will improve the lives of underserved PAD patients (e.g., CLTI patients, older adults encountering digital exclusion, some ethnic minority or socioeconomically deprived communities). It addresses unmet care needs, key evidence gaps and research priorities[16,36]. It will improve wellbeing, physical activity, prevent deaths,

reduce lower limb amputations and other cardiovascular complication rates. It aligns with the DHSC and government's focus on community/primary care[51,53]. This research will shape national/international community rehabilitation practice and PAD guidance[36], NHS care and this clinical context internationally. By reducing the need for more intensive treatments, emergency attendances and admissions, this research may generate significant NHS cost savings. This professorship will transform my career and be a springboard to international leadership. It increases research capacity in underrepresented research areas/disciplines as all postdoctoral fellows and PhD students are likely to be nurses or AHPs. Opportunities for doctoral supervision experience, funding application preparation and leadership training will develop these researchers to independence.

Project Management

I will be responsible for overall research delivery on time and within budget. I will monitor progress and expenses against milestones and report to the funder. I will meet my research team weekly. The research inclusion and public involvement officer will work with me to deliver workstream 1, first post-doctoral researcher will lead on workstream 2 and 5, the second post-doctoral researcher will lead on workstream 3 and, with collaborators, the doctoral student will lead on elements of workstream 4. These researchers will complete protocol development, governance approvals, research delivery and report writing. They will lead funding applications, trial set-up/commencement, with one or more possibly moving to work on them full-time.

Initially, I will meet my host mentor weekly then monthly. My professorship steering committee will meet twice every year. I will work with Professor Busse and colleagues at the Cardiff University Centre for Trials Research, who will support trial design, funding application development and trial delivery. The research host and sponsor will be City St George's who will ensure all contractual, legal, financial and regulatory obligations are achieved.

Research governance

Research will be conducted in accordance with relevant ethical, regulatory and legal requirements, including Good Clinical Practice, UK Policy Framework for Health and Social Care Research and data protection laws. Research governance approvals will be obtained from City St George's and NHS Research Ethics Committees/Health Research Authority, as required.

Success criteria

- Successful recruitment to all posts.
- Diverse patient, provider involvement, participation and engagement.
- Reaching workstream milestones (without significant delays).
- Building research capacity in underrepresented disciplines.

Barriers:

- Research inclusion (likelihood: low; impact: high): I will employ a dedicated co-ordinator, regularly monitor diversity and identify ways to maximise inclusion.
- Staff recruitment (likelihood: low; impact: high): The topic and duration of funding mean these posts are attractive. I will promote the roles via my networks.
- Research set-up/management (likelihood: low; impact: high): I achieve favourable ethical opinions via City St George's, IRAS and NIHR Coordinated Systems (HRA/NHS).
- Securing external funding (likelihood: medium; impact: high): I am experienced in securing funding for rehabilitation research. A PGfAR stage 2 application including workstream 3 and trial is already under consideration.

7. Patient & Public Involvement

Please describe how patients/service users, carers and the public have been involved in developing this proposal

The public have framed this professorship so that inclusion is at its core. Regularly, over two years, I discussed my research with diverse PAD patients and their carers in coproduction meetings that have shaped the focus, research questions, and plan. Nineteen people with lived experience of PAD, including people from Black, African Caribbean, South Asian, and Eastern European communities and those with CLTI shared their experiences of living with PAD and their treatment. To reach people from underserved communities who did not want to join a group or those experiencing digital exclusion, I talked to them individually via telephone or in-person.

They discussed the barriers to receiving rehabilitation in hospital (e.g. time, travel) and how they wanted to receive co-ordinated and responsive rehabilitation/support locally. They described how this research was 'much needed' as this part of NHS care was 'unacceptable and burdensome' and they asked for 'better access' to rehabilitation and ways to help them look after their condition and symptoms independently.

Patients with CLTI described having treatment in hospital and then ***'being left in a vacuum at home'***. They welcomed sessions with a provider to help them ***'make the most of their lives'.*** People with IC told me that home-based walking exercise was often ***'overlooked as a treatment'***, but that specific walking advice with support was needed. They liked the content and format of MOSAIC1 and commented that having this intervention provided in the community with ***'enough time with the therapist'*** was key to helping them walk more. They didn't mind who treated them '***just as long as they were well trained and supervised'***.

All advisors with lived experience of PAD recognised that becoming more active was important and wanted to explore mhealth using activity trackers, although some were nervous about this being the only treatment option. They were worried about ***'everything going online'*** and suggested developing an intervention that was easy to use, inclusive and accessible to everyone. This advice changed the focus of this workstream***.*** The public advisors helped identify research questions and develop the research plan. Three advisors (two from ethnic communities) read the public summary.

Please describe the ways in which patients/service users, carers and the public will be actively involved in the proposed research, including any training and support provided

My public involvement strategy aligns with the NIHR 2025-2030 strategic commitments and six UK Public Involvement Standards, ensuring valuable, sustainable relationships, quality and consistency[99,100].

Standards 1-3: Our advisory group includes men, women and members of underrepresented populations (n=9). I will seek further representation from underserved populations and diverse geographical locations by continuing to work with patient and community groups. I will employ a dedicated research inclusion and public involvement officer to plan, support and liaise with our advisors and community groups, co-ordinate our regular meetings and ad hoc activities. I will be responsible for oversight, planning and reporting of involvement, impact, and diversity in involvement, participation and engagement. Research inclusion and cultural competency training will be offered to public advisors.

Standards 4-6: My team will meet with our public advisors every 6-months to discuss workstream progress, and work with them in between on focused activities. We will balance meaningful involvement with the risk of over-burdening them. Digital exclusion will be mitigated by providing iPads, data packages and technical support, if needed, or if a member is not able to join the larger online meeting, they will be sent materials and can speak individually either remotely or via telephone. Advisors will be offered involvement in elements of workstreams that particularly interest them and invited to monthly research management meetings. Planned activities include providing feedback on public facing materials such as websites, reviewing study processes to enhance acceptability and experience of participation, promoting public awareness of the study, reaching hidden communities and dissemination. We will fund our public advisors to co-present at conferences and become co-investigators on funding applications.

Standard 6: Public involvement will be monitored, reviewed and reported at research management meetings, with public advisors providing feedback, reviewing and agreeing reported impact. This will enable reflection on our processes to improve advisor experience and explore quality and impact. We will report our public involvement in research using the Guidance for Reporting Involvement of Patients and the Public 2-short-form checklist[79]. All involvement activity and expenses will be reimbursed in line with NIHR policy.

If it is considered not appropriate and meaningful to actively involve patients/service users, carers and the public in your proposed research, please justify why
Not applicable.

8. Leadership & Development and Research Support

Proposed leadership and development programme

In 2017, I completed the King's College London emerging leader programme and, in 2021, the St George's, future leaders programme. Tangible benefits of these programmes have been greater national and international research collaborations and research leadership. My leadership development is continuing in health service research, community rehabilitation service delivery, capacity development and management and leadership at an international level.

Using Vitae's Researcher Development Framework, I have developed a leadership development programme, focusing on:

- ***Transition to senior strategic leadership and personal effectiveness: choosing commitments wisely (Domain B, C)***
- ***Maximising international networks, engagement and impact (Domain D)***
- ***Consolidating and expanding knowledge (Domain A).***
- ***Sustainable research capacity development (Domain D)***

This professorship will be a career step-change into international strategic research leadership aligned with NIHR priorities[16]. It provides senior leadership training, development opportunities to work with service planners and policy makers and the opportunity to foster exciting new collaborations through research visits and networking opportunities.

Over the next 5 years I will:

Amplify my senior strategic leadership and personal effectiveness as an international rehabilitation research leader who leads and influences others to bring about change across boundaries, with the **NIHR Future Focused Leadership Programme** (in year 1).

To consolidate my strong and dynamic research leadership, I will be mentored by Professors Mallen (primary care), Prof Busse (clinical trials, rehabilitation research), Prof Modarai (vascular) and Prof Jones (co-design, self-management, implementation). My personal development will be further supported by quarterly remote/in-person **leadership coaching** with Dr Carol Daniels, a coach with extensive experience of developing female academic leaders, throughout the 5-year professorship. The sessions will build personal effectiveness and leadership capability in key areas e.g. nurturing and developing others; influencing, mobilising individuals and groups; prioritisation and self-management; delegation and execution. This will enable me to transition to a more senior role towards the end of the professorship and understand the leadership needs of others in my teams.

International engagement, influence and impact: Leading at the level of NIHR professor will require me to engage and influence many different individuals and groups nationally and internationally. I plan to visit two leading PAD researchers: Prof Mary McDermott (Northwestern University) and Prof Andrew Gardner (Penn State College) in the USA (year 2). These researchers lead groundbreaking inclusive community-based PAD studies and transformed clinical services by introducing targeted interventions for underserved groups. Coproducing research and learning from them will directly inform my inclusive PAD research programme and spark new international collaborations.

To reinforce my knowledge and ensure that I remain abreast of current developments in remote physical activity monitoring and mhealth, I will join the Faculty of Physical Education and Sport, Charles University, Prague, Czech Republic (in year 1). Dr Tomas Vetrosky, a new collaborator (workstream 4), is a leading European researcher in inclusive mHealth and physical activity trials in people with long-term conditions. Collaboration with his centre will include further learning on ecological momentary analysis, digital regulations and will spearhead a new European research collaboration.

To further my international research and leadership profile, and networks, I will present my research at the Society for Vascular Surgery, World Physiotherapy, World Congress on Rehabilitation, International Primary Care and Accelerating Health Equity Annual Conferences.

Building knowledge and skills: To complement my existing clinical research skills, I will work alongside and visit Cardiff University Centre for Trials Research (mentor: Busse) to learn about innovative rehabilitation trial designs and identifying and developing inclusive, meaningful outcomes for community rehabilitation trials (visit: Year 3). I will apply for City St George's delivered short courses on responsible conduct in research (governance, ethics and integrity) and advanced project and financial management to ensure that I remain up-to-date with regulatory requirements throughout the 5-year professorship. I will also schedule the

- 4-day media and science communication masterclass at Bristol University to help me maximise the dissemination and impact of my research (year 2).
- 5-day course in advanced clinical trial design (adaptive, platform, sustainable trials) at Birmingham University to develop my skills in complex, innovative trial design and delivery (in year 3).
- 5-day knowledge into Action course at Oxford University to optimise adoption and implementation of my research (year 4).

In pioneering the shift of PAD rehabilitation into community services, I will participate in one-to-one coaching with Dr Catherine Huffington, Director of Health improvement Southwest London ICB to strengthen my knowledge of planning and commissioning community services.

Sustainable research capacity development: I will build a sustainable inclusive strategy for rehabilitation research capacity development within my Professorship that expands my current and continuing researcher capability development work and future proofs the NIHR-supported Community Rehabilitation Research Incubator hub that I lead. I will develop the next generation of postdoctoral and doctoral AHPs to become community rehabilitation research leaders, building capacity in underrepresented professions through their own successful personal fellowship and funding applications.

Career intentions

This Professorship will give me an extraordinary opportunity to deliver exceptional and much-needed community rehabilitation research to transform the care of PAD patients. City St George's has committed to continuing to support me and my research beyond this award through access to training opportunities for current and new researchers in my group and providing administrative and infrastructural support. At the end of this award, my goals are to

- Be leading a multidisciplinary, collaborative team who will maximise inclusive research and support the delivery and mobilisation of my international, high impact research.
- Have developed inclusive PAD community interventions, secured further funding and developed strategies for implementation to transform community services and maximise impact.
- Move from national lead in PAD research and knowledge mobilisation in practice to enabling high quality methodological research to be done at a pace.
- Increase my national and international profile, enabling me to maximise the opportunities to contribute to international guideline groups, grant panels (e.g. NIHR HTA), and to speak at high-profile meetings where impact is greatest.
- Have established a sustainable collaborative strategy and network to nurture the future generations of community rehabilitation researchers at all career stages.
- Be a strong candidate for NIHR Senior Investigator status.

RESEARCH SUPPORT 1

Name	Professor Fiona Jones

Institution	City St George's, University of London

Position	Professor of Rehabilitation Research

Description of support provided
Professor Fiona Jones MBE is a Professor of Rehabilitation Research with an international reputation in co-design, evaluation and implementation of self-management and rehabilitation interventions. She launched the successful spin out community interest company Bridges Self-management (Bridges) in 2013, which employs 20 staff including people with long-term conditions. Bridges has trained over 500 healthcare teams across the UK and internationally, to integrate evidence-based self-management support and been included into many large NIHR funded projects across different long-term conditions. Her multi-professional research group comprises doctors, AHPs, nurses, methodologists and implementation scientists. She champions AHP research capacity as a NIHR Advanced Fellowship panel member, ARC Implementation Science deputy theme lead and founding member of the Community Rehabilitation Research Incubator hub. Professor Jones has supported my advancement since joining City St George's, and is a source of honest and open feedback. She will provide specific research leadership mentorship and wider career advice. She will support the experience-based co-design and implementation aspects of the planned work as well as advise on public involvement and engagement. She will guide my objectives for research inclusion and authentic and diverse public involvement and engagement. Her support will ensure that my science remains rigorous and relevant whilst I develop leadership skills and expertise. She will be a critical role model and, together with Professor Busse, guide me as I develop as a strong female rehabilitation research leader. Initially we will meet weekly and then monthly, and Prof Jones will chair the biannual professorial steering committee.

RESEARCH SUPPORT 2

Name	Professor Christian Mallen

Institution	Keele University

Position	Professor of Primary Care, Director, NIHR School for Primary Care Research

Description of support provided
Christian Mallen is a past NIHR Research Professor, NIHR Senior Investigator and Director of the NIHR School for Primary Care Research (SPCR). He is an academic general practitioner with a portfolio of research focusing on improving the diagnosis and management of musculoskeletal and rheumatological disorders in community settings. He has a strong track record in multi-disciplinary working and supporting career development, as past academic capacity development lead for the NIHR SPCR and director of the Wellcome PhD Programme for Primary Care. His research group is multi-professional including clinicians (physiotherapist, nurses, pharmacists, psychologists and GPs) and methodological experts (clinical trials, epidemiology, data synthesis and social sciences). Professor Mallen's mentorship will fulfil three essential roles in my future career development. Firstly, as a leading international expert in primary and community care, where my research is focused, he will ensure my research remains relevant to the needs of patients and primary and community care practitioners and is

developed to be applicable in that setting.

Secondly, his experiences of developing an internationally renowned multidisciplinary programme of research and his leadership role with the NIHR SPCR will provide a unique perspective on academic training and capacity building which will support me to build a strong cohort of multi-disciplinary researchers.

Thirdly, he will provide opportunities for formal collaboration with world-leading general practice researchers, for example, Professor Paul Little, University of Southampton.

Professor Mallen will join the professorship steering committee that meets twice every year and be available for ad hoc advice via remote meetings or email.

RESEARCH SUPPORT 3

Name	Professor Monica Busse

Institution	Cardiff University

Position	Director of the Health and Care Research Wales Faculty

Description of support provided
Professor Monica Busse is an applied health researcher and trial methodologist based in the Cardiff University Centre for Trials Research (CTR), Director of the Health and Care Research Wales Faculty, and current Health and Care Research Wales Senior Research Leader (Senior Investigator equivalent). Her research seeks to enable people living with chronic health conditions to sustain an active lifestyle while staying connected and engaged in meaningful aspects of their lives. In CTR, she is Chief Investigator (or co-CI) of a suite of innovative NIHR and charity/government funded rehabilitation trials. Prof Busse is an accredited executive coach with an interest in promoting research careers and developing future research leaders and is ideally placed to mentor me as I move through this professorship and programme of work. Specifically, I will draw on her experience, knowledge and insight of developing an internationally renowned multidisciplinary research programme when • designing and delivering innovative trials. Prof Busse has agreed to host a research visit to the CTR. • developing strategy and building a strong culture of inclusion and a positive working environment. • planning and enabling research capacity development of my team and others in my broader network. Together with Professor Jones, she will act as a role model and nurture my development as a strong female research leader. Prof Busse is a vital member of my professorship steering committee that will meet twice every year. Prof Busse will also be available for ad hoc advice via remote meetings or email.

RESEARCH SUPPORT 4

Name	Professor Bijan Modarai

Institution	King's College London

Position	Professor of Vascular Surgery

Description of support provided
Professor Bijan Modarai is the King's Chair in Vascular Surgery, a past British Heart Foundation Senior Clinical Research Fellow, current President of the British Society for Endovascular Therapy and leads the King's Health Partners Cardiovascular Clinical Academic grouping. He is an academic vascular surgeon whose work focusses on improving diagnosis and management of people with vascular disease by developing novel diagnostics and advanced therapeutics. He is an advocate of "team science", leading translational programmes of vascular research that engages multi-disciplinary collaborative teams, including clinicians (vascular scientists, physiotherapists, nurses) and vascular biologists, data scientists, statisticians, imaging experts and radiation biologists. He is an internationally recognised clinical-academic with an extensive network of international collaborations that he has leveraged for impactful outputs. I will build on our existing relationship, cultivated from collaborating with him on several research projects since 2010. I am honoured that he has agreed to provide mentorship through my transition to NIHR professorship. Specifically, his mentorship will focus on: • my engagement with international collaborators to highlight areas of unmet need and establish vascular research. • emerging areas and the wider context of management of vascular patients. • through his extensive international collaborative network, he will connect me with vascular surgeons who lead highly impactful research and broker new research collaborations e.g., Kevin Mani Professor of Vascular Surgery, Uppsala, Sweden. He will contribute to my professorship steering committee that meets twice every year and be available for ad hoc remote advice and direction on both the science and my career trajectory.

Collaborations
My Professorship will extend my successful partnerships and create important new collaborations. I will visit and forge new ***international collaborations*** with: **Dr Tomas Vetrovsky,** Faculty of Physical Education and Sport, Charles University, Prague, Czech Republic. Dr Vetrosky is a leading European researcher in inclusive mHealth/JITAI and physical activity trials in older people with long-term conditions. I will visit Dr Tomas Vetrosky's centre in year 1. His knowledge from the WATCHFUL and ENERGISE trials that evaluated JITAI and inclusive engagement and participation strategies in older adults with long-term conditions will be critical for workstream 4[101,102]. I will consolidate and extend my learning on innovative mhealth interventions and research methods e.g., ecological momentary analysis, digital regulations. He is a significant collaborator and will act as an external PhD supervisor. This will be a new European research collaboration. I plan to visit and collaborate with two leading PAD researchers in the USA: **Professor Mary McDermott,** Northwestern University and Memorial Hospital, Chigago and **Professor Andrew Gardner,** Department of Physical Medicine and Rehabilitation, Penn State, in year 2. These researchers lead groundbreaking inclusive community-based PAD studies and have transformed clinical services by introducing targeted interventions for underserved groups. Professor McDermott and her team have undertaken the flagship HONOR and GOALs trials into home-based PAD exercise/mhealth interventions[103,104]. Professor Gardner investigates PAD underrepresented groups e.g., people with PAD/diabetes and ethnic minority communities. By working alongside these teams, I will gain an in-depth understanding of inclusive trial design for underserved PAD groups and integrating these into equitable clinical services. This will inform all workstreams and will spark other avenues for international collaboration. I will build on ***existing international collaborations*** with clinicians, methodologists and patients in the vascular and rehabilitation communities: I will continue to link with the Professor Janet Peacock, Professor of Epidemiology and Biomedical Data Science and Professor Jeremiah Brown, Centre for Implementation Science at Dartmouth Geisel School of Medicine, USA to ensure my interventions and strategies are rigorously evaluated, implementable and

sustainable in real-world settings.

I will continue to collaborate with ***UK clinicians and researchers***: Professor Athanasios Saratzis, Professor of Vascular Surgery and Professor Charlotte Edwardson, Professor of Physical activity, sedentary behaviour and health, Leicester University; Dr Louise Hull, Implementation Scientist at King's College London, all co-investigators on my PGfAR application and Professor Monica Busse, Cardiff University. I will maintain my strong relationship with clinical and methodological colleagues at City St George's, University of London: Professor Michael Ussher, Dr Mandeep Sekhon, health psychologists/behaviour change experts, Professor Tess Harris, Professor of Primary Care Research and physical activity triallist.

Partnering with research inclusion experts: Having established relationships with UK vascular patient organisations to maximise research inclusion (e.g., national vascular public advisory group), I will consolidate my relationships with communities that work with PAD underserved groups: the Centre for Ethnic health research and the national Romanian and Eastern European Hub. I will extend my links to other organisations such as Caribbean & African Health Network, an organisation which seeks to address health inequalities disadvantaging people of Black Caribbean and African heritage.

To ensure that my research is relevant to community-based needs, is user driven and appropriate for widespread roll-out by service planners, I will meet regularly with Zoe Lelliott, current NIHR ARC South London Deputy Director and former CEO of the South London HIN. I will keep abreast of the challenges encountered by service planners by continuing to partner with the Southwest London ICB research group and develop my relationship with Dr Catherine Heffernan, Director of Health Improvement and her primary and community services commissioning team via visits and one-to-one meetings.

Host Organisation Support Statement
The following statement of support must be completed by the lead signatories from your **host institution** and **partner organisation**. In order to contribute to the application, these individuals must be invited to do so via the 'Participants and Signatories' section of the form.

Statement of support

City St George's, University of London together with St George's University Hospital NHS Foundation Trust (SGUHFT) strongly support Professor Bearne's NIHR Professorship application.

Since joining City St George's, Professor Bearne has a rapidly accelerating career trajectory and her research plans are ideally matched to the NIHR Professorship aims, capitalising on her successful research career. Lindsay is committed to undertaking applied research that will improve patient health and care. She has successfully conducted multi-site trials that have been presented at international conferences, published in high profile journals (e.g., JAMA) and informed national guidelines (e.g., NICE). Her competitively awarded NIHR secondment recognises her ability to develop and shape national strategy to maximise research impact.

Professor Bearne leads effective, multidisciplinary groups to address the challenge of supporting people with long-term conditions to live well. Her success building and sustaining national collaborations is due to her extensive research knowledge, skills and experience, innovative and original ideas, personal qualities, and ability to deliver projects. Her research into community rehabilitation for long-term conditions, particularly peripheral arterial disease, demonstrates that she is adept at identifying clinically important research questions and building teams that have the methodological and subject expertise to answer them. There are very few allied health professionals (AHPs) at Professor Bearne's level who possess this combination of original and relevant research ideas, a highly collaborative approach and leadership ability. This Professorship will enable Professor Bearne's research to flourish, increase its global reach and maximise its impact.

Organisational support

Reflecting her impressive track record, research and teaching achievements, and excellent references, Professor Bearne was recruited to City St Georges in 2021. Recognising her abilities and strategic importance, we have supported her through training and capacity development awards to secure >£8 million collaborative funding, including obtaining funding for and leading the NIHR-supported Community Rehabilitation Research incubator

hub.

Professor Bearne was selected through a competitive internal process to be supported to apply for this award, reflecting her steep upward career trajectory and her potential as a future leader. This application is of immense importance both addressing critical questions and for the development of

Professor Bearne. It expands her research in a crucial new direction which we believe will propel her from national success to be an international leader.

City St George's Tooting is in a unique position to support Professor Bearne's research. It is completely dedicated to medicine, healthcare, and science, sharing its site with SGUHFT. The Trust is one of London's biggest hospital Trusts, providing services to 3.5 million people with >9000 staff. The integration of patients, healthcare workers, students and scientists, promotes a dynamic learning environment. The Population Health Research Institute (PHRI), where Professor Bearne is based, is at the heart of our research activity. PHRI is a multidisciplinary research unit with an outstanding record of conducting high quality research to underpin population health improvements. The PHRI comprises ~60 staff, with 19 principal investigators. Its 2022-2030 Strategy includes influencing national and global health policy; research inclusion and use of co-design; and focusing on delivering research with real world impact. In the last Research Excellence Framework, St George's was ranked 8th for impact. 91% of PHRI outputs were ranked as internationally excellent or world leading. City St George's has ARC South London membership and actively collaborates with Southwest London Integrated Care Board which further facilitates co-production of impactful research of regional and national importance. Professor Bearne will contribute to the current and future success of PHRI, and PHRI will provide excellent strategic and practical support (internationally renowned methodologists in evidence synthesis, statistics, qualitative and participatory methods).

City St George's is fully committed to Equality, Diversity and Inclusion employment practices, as emphasized in its Strategic Vision and overseen by the Dean for Equality, Diversity and Inclusion. We are committed to the VITAE Concordat to support the career development of researchers and to the principles of the European HR Excellence In Research Award. City St George's has held an Athena Swan Silver Award since 2018 (one of only 16 universities) and is a Stonewall Diversity Champion and Disability Confident Employer.

Contribution to service

Professor Bearne holds an SGUHFT honorary contract to support AHP clinical service development and research capacity building. She provides research training, supports service improvement initiatives, development of clinical research projects and NIHR fellowship applications. With Professor Bearne's support, the number of successful SGUHFT AHP fellowships has grown in the past 3 years (currently 3 PCAFs, 3 DCAFs, 1 ACAF). Professor Bearne will dedicate one session/week to support AHP research capacity development and service improvement. This will directly influence clinical services and patient outcomes through supporting clinicians to become critical consumers of research, championing evidence-based practice and a positive research culture, establishing local research networks and acting as an AHP research role model. This will maintain Professor Bearne's understanding of key clinical issues, develop her team by offering supervision opportunities and expand the cohort of SGUHFT clinical academics. The University will release Professor Bearne from all leadership positions and teaching responsibilities. She will discontinue her NIHR secondment.

The University/Trust has committed to an exceptional support package for Professor Bearne. This includes a range of new posts, studentships, leadership and development of both Professor Bearne and her team to boost profile and build current and future capacity.

National and international level support

Professor Bearne is an academic physiotherapist with an internationally recognised research profile. She is acknowledged as a national leader in rehabilitation of long-term conditions, and an emerging international leader in community rehabilitation. She has established, through physiotherapy and vascular interest groups and the Community Rehabilitation Research Incubator Hub, a wide UK network. We support her plans to attend international conferences and to build new collaborations in the Czech Republic and USA.

Lindsay Bearne is a major talent with an established record in grant income and research outputs. Our partnership's backing will synergise with the award of this Professorship to support Professor Bearne to rise to the challenge of becoming an international leader and will provide a step-change to her development.

For the recycling of the applicant's salary, please state how this will be used to support the nominee.

For Professor Bearne to gain maximal leverage from the NIHR Research Professorship whilst still leading her existing portfolio of research, we have agreed to provide significant support. Release of 100% FTE salary commitment by City St George's to Professor Bearne for the 5 years of the NIHR Professorship will enable this to be recycled in five ways:

- **New Lectureship** (1.0 WTE; £399,564): To support and build the academic base for AHP and rehabilitation research. In addition, they would take on Professor Bearne's teaching and supervisory responsibilities, for example MSc students in pre-registration Physiotherapy and Advanced Clinical Practice teaching and project supervision.
- **Additional PhD studentship** plus fees and training budget (£92,333), aimed at an underrepresented discipline e.g., nurse or allied health professional with specialist interest in community rehabilitation, physical activity or vascular conditions.
- **Training support for the NIHR funded PhD studentship** (£7000). The university will fund training and research development for the NIHR funded PhD student to maximise research capacity building in underrepresented professions and optimise the NIHR Professorship investment.
- **New research inclusion and public involvement officer** (0.2 FTE: £58,377). To facilitate outreach and engagement with PAD underserved communities and co-ordinate all public involvement, participation and engagement activities for all workstreams.
- **New research administrator** (0.3 FTE: £87,566). To provide diary, meeting, travel, expense and research management support primarily to Professor Bearne and the new Lecturer, but also research team members more generally as needed for the efficient and effective running of all workstreams.

These costs are substantially greater than her 100% FTE professorial salary so the shortfall in the funding required for these posts will be met by the University.

These appointments will substantially reduce the burden of administration, teaching, supervision and meetings, allowing Professor Bearne to focus on research design, delivery and research funding; mentoring and supervising early career researchers; building her national and international leadership and academic networks; and maximising the impact of her research on practice and policy, to benefit patients, practitioners and the NHS.

Please state what additional support will be given to the nominee over the short, medium and long term.

City St George's and SGUHFT have a long track record of partnership in research, innovation, and supporting training and development of applied researchers and clinical academics. We are committed to providing strong support to ensure the impact of Professor Bearne's research is maximised. We will commit substantially more than 100% FTE of Professor Bearne's salary.

Short-term (1-3 years)

Support in leadership and development

The University will fund personal coaching, observation and 360-degree appraisal for Professor Bearne (£15,860), and attendance at the City St George's leadership programme and short courses on responsible research conduct (governance, ethics and integrity) and advance project and financial management as described in the Leadership development section.

Professor Bearne's team members will have access to

- City St George's researcher development programme. This provides short courses in key transferable skills mapped against the Vitae Researcher Development Framework, addressing research practice, integrity, ethics, time-management, public engagement and core disciplines (statistics, methods, planning, critical appraisal). Estimated cost/researcher; £1000 (total: £5000).
- City St George's Postgraduate Certificate in Clinical Research (£3250/PhD student, total: £6500) or relevant modules from the Master's in Clinical Research, as appropriate (£7,000 for up to two modules per postdoctoral fellow; total: £14,000).
- The South London ARC Research Leadership Academy training programme. This structured 6-month leadership programme for early career researcher includes: identifying your research vision, leadership qualities development, research management, driving research quality and maximising impact, involving of diverse and minoritised groups and individual coaching and mentorship
- The new lecturer will be supported to complete the universities educational training programme (training, supervision, registration fee) to achieve Fellowship of the Higher Education Academy (estimated cost; £3000)
- 3-yearly Good Clinical Practice Training.

Mentorship

Professor Bearne will receive high-quality mentorship at City St George's (Professor Jones; see mentorship section) and in addition she will be offered senior departmental support and mentorship within PHRI and via The Centre for Healthcare Innovation Research.

IT infrastructure and support

Each additonal member of Professor Bearne's team (lecturer, PhD student, research inclusion and public involvement officer) will be provided with dedicated office space and will receive a light-weight high-performance laptop to enable flexible working (£3000). The University will provide 5TB long-term (up to 10 years) **data storage** for each research team member.

Support in research management

Through recycling of 100% FTE of Professor Bearne's salary, and additional contribution by the University, new posts will be appointed to support her programme of work.

A **research inclusion and public involvement officer** will be employed to cover outreach and engagement with PAD underserved communities and co-ordinate all public advisory group, inclusive participation and engagement activities for all workstreams.

An **administrator** will provide support to Professor Bearne, diary and meeting management, co-ordinate the appointments and line management of Professor Bearne's team, and coordinate/support on-going and planned research, including day-to-day monitoring of and the wider team, co-ordinating meetings and correspondence with coinvestigators.

In addition, the University will provide **contracting and financial management** support, and **research management** support, through the Joint Research and Enterprise Service, to oversee regulatory and reporting requirements.

Medium term (4-5 years)

We will continue to support Professor Bearne as outlined above, plus

- We intend to support the launch of a new **Centre for Community Rehabilitation Research**, with Professor Bearne as Director. This will build on her research portfolio and leadership of the NIHR-supported Community Rehabilitation Research hub. It will focus on developing research capacity in underrepresented professions. The ambition of this Centre aligns with the strategic direction of our newly merged university; to foster clinical academic researchers and do translational, impact-oriented and clinically engaged science. It will raise Professor Bearne's and her group's profile beyond healthcare and promote cross boundary collaboration with specialists based in social care and public health**.**
- We will provide dedicated training for Professor Bearne's research team to develop their independent research programme through personal fellowship applications

Long-term (5+years)

Beyond the NIHR Professorship we will continue to support Professor Bearne and the Centre for Community Rehabilitation Research to build a critical mass of researchers in community rehabilitation. We plan to:

- Support the mid-career researchers appointed to this research through further development, training and support with the aim of providing long-term job security and appointments.
- Continue to support this research programme and Centre through continued investment in research infrastructure and research management support.

9. Detailed Budget

The finance section should provide a breakdown of costs associated with undertaking the research as described in the proposal.

10. Uploads

(References - Lindsay Bearne Research Professor Reference List.pdf) is included as an appendix within this file.

(Figures / Tables - Supplementary Information.pdf) is included as an appendix within this file.

(Research Timetable - Lindsay Bearne Gantt Chart .pdf) is included as an appendix within this file.

(Peer review publications's summary - Lindsay Bearne Research Professorships publication record.pdf) is included as an appendix within this file.

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11. Participants and Signatories

Participants

Participants must complete their actions on your application prior to submission; signatories must also approve your application after you select the 'submit' option but BEFORE the application submission deadline.

You are required to supply the names and email addresses (if not already registered on the ARAMIS application system) of the individuals who will be undertaking 'participant' roles as part of your application. Everyone named in this section will be acting as a 'participant' to your application and will need to agree to be part of this application.

By confirming participation, participants are acknowledging their involvement and input into this application and agree to be involved in it before it is submitted. You must ensure all participants are happy for your application to be submitted before submitting it on the online system.

Details of the individuals who will be required to approve your application after submission (signatories) should be entered in the 'Participants and Signatories' section. Please note that it is acceptable for a Head of Department to also act as a research support role. If this is the case, they will need to be included in this section for both roles.

Participants must confirm their participation on your application before you will be able to press the submit button. Please see the 'Application Submission Process Flow Diagram' in the guidance notes for further information.

Research Support

Research Support Participant

Title	Professor
First Name	Monica
Last Name	Busse
Position	Director Mind Brain Neuroscience Trials

Research Support Participant

Title	Professor
First Name	Fiona
Last Name	Jones
Position	Professor Rehabilitation Research

Research Support Participant

Title	Professor
First Name	Christian
Last Name	Mallen
Position	Professor of General Practice

Research Support Participant

Title	Professor
First Name	Bijan
Last Name	Modarai
Position	

Signatories

Participants must complete their actions on your application prior to submission; signatories must also approve your application after you select the 'submit' option but BEFORE the application submission deadline.

You are required to supply the names and email addresses (if not already registered on the ARAMIS application system) of the individuals who will be 'signing off' your application. Once their contact details have been entered below, the signatories will be invited to log into the system and confirm their participation. Details of what is required and expected of each of the roles can be found below.

Once participation is confirmed, the Heads of Departments must provide the required statements in the 'Leadership & Development and Research Support' section. This statement can be completed independently whilst you (the applicant) work on the rest of the application.

Signatories must confirm their participation and complete their relevant sections before you will be able to press the submit button.

The signatories must also approve the application after you have selected the 'SUBMIT' option but **BEFORE** the application submission deadline. Please see the 'Application Submission Process Flow Diagram' for further information.

The final signatory approval will result in the application being fully submitted to the NIHR. All parties (applicant, participants and signatories) will be notified of this via an automated system generated email. **NIHR will not accept any applications unless fully approved by your signatories prior to the 1pm deadline on Wednesday 11th December 2024.**

Lead Signatory of Host Organisation

Lead Signatory of Host Organisation

Title	Professor
First Name	Rachel
Last Name	Allen
Position	Institute Director

Representative of Partner Organisation

Representative of Partner Organisation

Title	Professor
First Name	Heather
Last Name	Jarman
Position	Group Director for Health and Care Professions Research

Administrative Authority or Finance Officer

Administrative Authority or FO Participant

Title	Mr
First Name	Adam
Last Name	Bales
Position	Research Funding Manager (Pre-Award)

12. Acknowledgement, Review and Submit

Conflict checks:
Please declare any conflicts or potential conflicts of interest that you may have in undertaking this research, including any relevant, non-personal & commercial interest that could be perceived as a conflict of interest.

Professor Bearne is seconded to the NIHR Central Co-ordinating Facility as a part time (0.4WTE) Senior Research Fellow in Knowledge Mobilisation.

Agreement to Terms and Conditions:

I confirm that I have secured all necessary licences and ethical approvals in relation to the research and will abide by the terms of those licences and approvals in carrying out the research.

I confirm that I have read the NIHR Carbon Reduction Guidelines and, where possible, taken steps to reduce the carbon emissions generated by this research.

I confirm that all material within this application, with the exception of sections that other participants are required to complete, is my own work and I have taken steps to ensure that no parts of the application have been inappropriately plagiarised.

I undertake to conduct the research according to such guidelines for good research practice as the Host Organisation may from time to time laydown according to the NIHR research governance framework.

In completing this, I declare that, to the best of my knowledge, the information provided in this application, is true, accurate and complete. I have read the Guidance Notes and agree to accept the process by which an application is assessed and agree to abide by the conditions under which an award may be granted.

☒Confirmed

Appendices

1. Lindsay Bearne Research Professor Reference List.pdf
2. Supplementary Information.pdf
3. Lindsay Bearne Gantt Chart .pdf
4. Lindsay Bearne Research Professorships publication record.pdf

Appendix #1: Lindsay Bearne Research Professor Reference List.pdf

1.Bearne. *JAMA* 2022;327(14):1344-1355.
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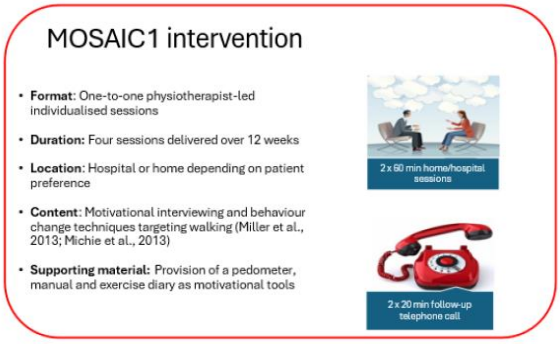
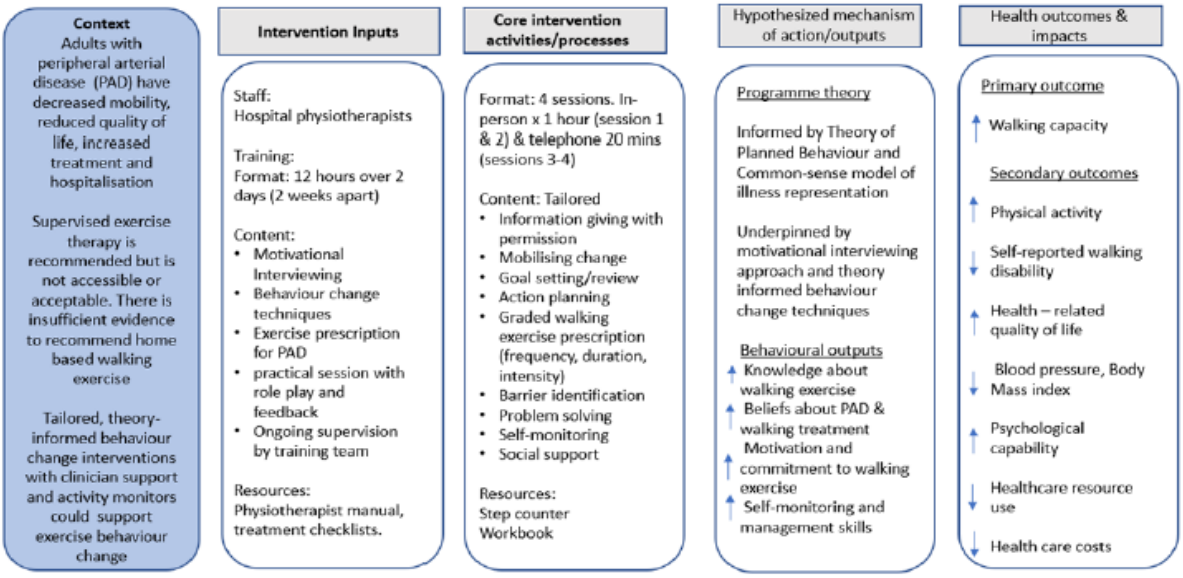
Appendix #2: Supplementary Information.pdf

Figure 1. Lindsay Bearne Research Professorship Programme Theory

Context	Inputs/activities	Outputs	Outcomes	Impact
Peripheral Arterial Disease: 3rd most common CVD condition 20% people over 70 years Disproportionally affects older adults, people with diabetes Black/African Caribbean, Eastern European, South Asian and socioeconomically deprived communities Treatment spans primary, community secondary and tertiary care. Patients want community care, such as supported self-management, rehabilitation and digital physical activity and exercise support yet it is not evidence based, inconsistently and inequitably provided	Inclusive and diverse national public involvement, participation and engagement research and implementation strategy Identification of relevant and important research questions Skills and experience in systematic reviews, intervention development qualitative and quantitative data generation and analysis Health service research design and delivery experience and skills Effective leadership and development of multidisciplinary research team Senior leadership training and development Development of early career researcher capability and capacity in under-represented professions	Co-designed supported self-management resources and provider training that is accessible and suitable for all CLTI patients Inclusive home-based walking exercise programme and provider training for IC patients Inclusive mhealth intervention to increase physical activity in all PAD patients PAD community rehabilitation implementation and evaluation strategy High quality publications, bespoke outputs for key audiences and impact case studies Community rehabilitation research professor with international reputation in senior leadership role Next generation of community rehabilitation researchers Three funding applications for definitive trials	More equitable access to and engagement with inclusive interventions for PAD (including mhealth) Evidence generated to improve community rehabilitation across the PAD symptom range Increased patient self-management knowledge, skills & confidence Increased physical activity , walking capacity, reduced co-morbidity and mortality. Internationally renowned research group in community rehabilitation Increased research capacity and capability in underserved professions	Improved community rehabilitation services for people with PAD across the symptom range Increased participation in daily activities, social participation, emotional well-being, quality of life across the PAD symptom range Increased in skills confidence and knowledge of PAD management for patients, carers, providers and planners Reduction in health and social care resource use Increased community rehabilitation High quality research community rehabilitation research that addresses critical

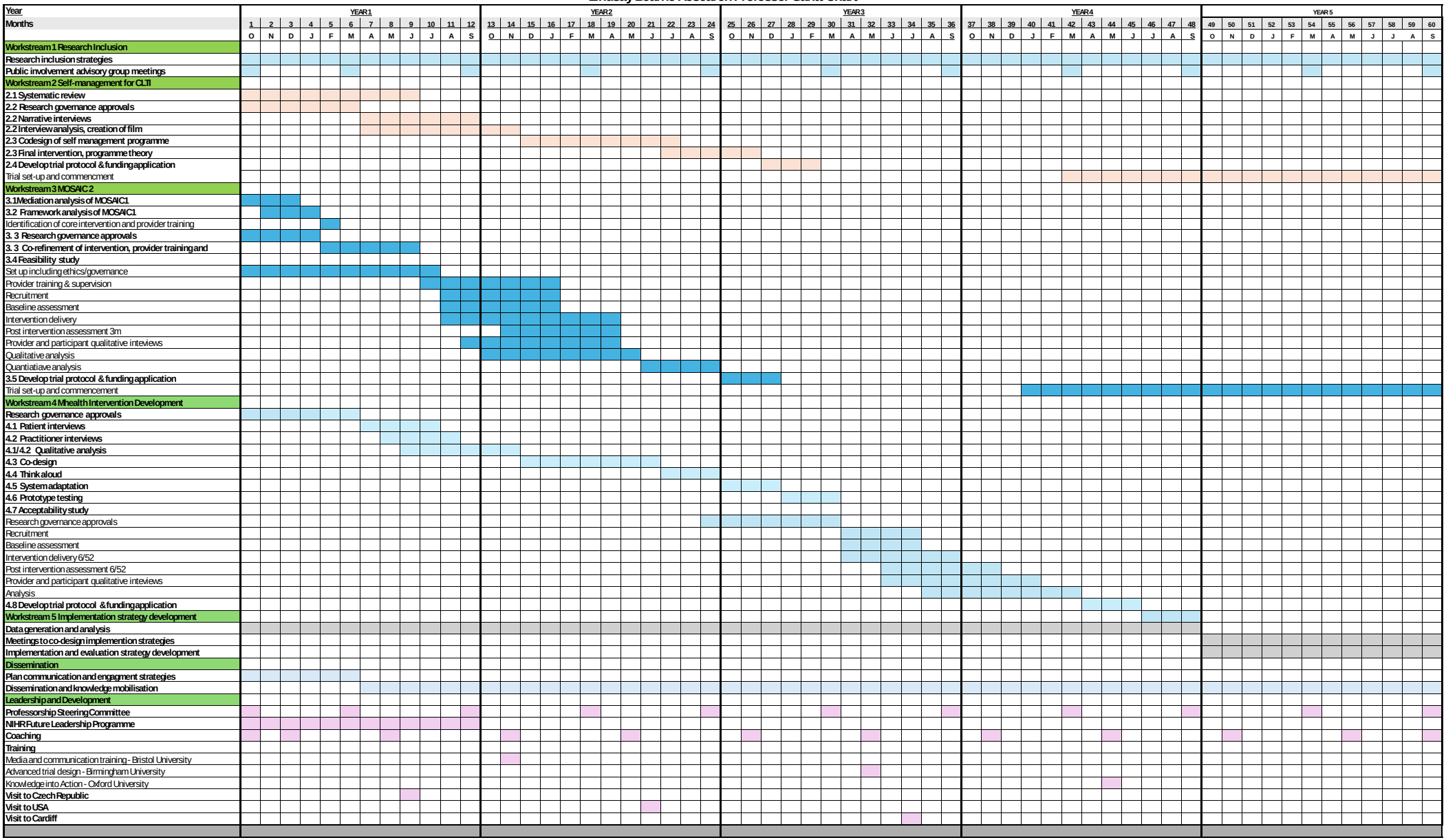
Figure 2. MOSAIC1 Intervention and Programme Theory





Appendix #3: Lindsay Bearne Gantt Chart .pdf

Lindsay Bearne Research Professor Gantt Chart



Appendix #4: Lindsay Bearne Research Professorships publication record.pdf

NIHR | National Institute for Health and Care Research

NIHR Research Professorships & NIHR Global Health Research Professorships Publication Record

Peer Reviewed Publication Summary

Please provide details of the most significant 20 original, peer-reviewed publications that would be of relevance to this application from the last 5 years. Please list in date order, with the most recent first. No additional rows may be added to this table. If more than 20 publications are listed in this section, only the first 20 will be taken into account. The panel assesses each publication on its merits, particularly in relation to patient benefit. Different disciplines vary in the journals that they use and journal impact factors will only form a part of the assessment.

I. Publication – please state the publication title.
II. Journal – please state the journal that the output was published in.
III. Year – please state the year published.
IV. Current Journal Impact factor – given that these change, please list the most current Impact factor for the journal listed.
V. Number of Citations – please list the number of times that the publication has been cited.
VI. Source of Citation Information – please state where you obtained the citation information from for each publication, e.g. Scopus, Google Scholar etc.
VII. Number of authors – please list the total number of authors associated with the publication.
VIII. Key Authorship Position – please state your authorship position for each publication, e.g. first, last or corresponding author.
IX. Interdisciplinary research – please choose either yes or no from the drop down list.
X. URL – please provide the web link to the journal (where available).

Other significant publications or outputs

Please provide details of any other significant publications, outputs or products that would be of relevance to this application, which provide evidence of influence and leadership in the nominee's field. Items in this section should include the following headings (where appropriate):

- Books and/or book chapters
- Papers in conference proceedings
- Research reports to government departments, charities, the voluntary sector, professional bodies, industry or commerce
- Monographs
- Intellectual Property (patents, patent applications or other forms of IP)

Other applied research outputs (new materials, software packages, images and devices, research derived from development, analysis and interpretation of bio-informatic databases, work published in non-print media, Cochrane reviews and NICE Guidelines).

Section 2: Nominee Curriculum Vitae

Overall Summary Tables

Publications:

Please provide the following information using Web of Science in the first instance, or Scopus thereafter. Please do not use any other system.

Overview	**Example**
Total number of Publications	118
Total number of Peer Reviewed Publications (full papers)	71
Total number of Senior Author Publications (first and last full papers)	37
Total number of Citations	1981
H index	21
Timescale this applies to	18 years (8.5 dedicated research years - see contextual factors)

Type of publication	**Number**
Journal Articles	71
Reviews	16
Book Chapters	5

Type of publication	Number
Books Edited	0
Conference Associated Publications	47
Editorials	2
Case Reports	0
Letters	1

Please state which system was used to obtain this information: Web of Science

Funding:

Type of funding	Total amount (£)
Current	8,798,032.00
Previous	2,868,334.00
Awaited	2,898,170.00

PhD Supervision:

- Number of PhD/Higher Degrees currently supervising: 3
- Number of PhD/Higher Degrees supervised to completion: 6

Section 2: Nominee Curriculum Vitae - Peer Reviewed Publication Summary

(Please list in date order with most recent first, the nominee's twenty most relevant significant peer reviewed original publications from the last five years)

	Publication	Journal	Year	Current Journal Impact Factor	No. of Citations	Source of Citation Information	No. authors	Key Authorship position	Interdisciplinary Research	URL/DOI
1	Effect of Behavior-Change Interventions on Daily Physical Activity in Patients with Intermittent Claudication: The OPTIMA Systematic Review and Meta-Analysis	European Journal of Preventive Cardiology	2024	8.4	0	Google Scholar	16	No	Yes	10.1093/eurjpc/zwae296
2	Participants' experiences and acceptability of a home-based walking exercise behaviour-change intervention (MOtivating Structure walking Activity in people with Intermittent Claudication (MOSAIC))	Physiotherapy	2024	3.01	1	Google Scholar	11	Senior	Yes	10.1016/j.physio.2023.09.002
3	Examining the Effectiveness of Social Media for the Dissemination of Research Evidence for Health and Social Care Practitioners: Systematic Review and Meta-Analysis	Journal Of Medical Internet Research	2024	7.08	0	Google Scholar	9	Senior	Yes	10.2196/51418
4	The effectiveness of social prescribing in the management of long-term conditions in community-based adults: A systematic review and meta-analysis	Clinical Rehabilitation	2024	3.04	1	Google Scholar	4	Second	Yes	10.1177/02692155241258903

5	Synthesis of guidance available for assessing methodological quality and grading of evidence from qualitative research to inform clinical recommendations: a systematic literature review	RMD Open	2024	5.1	0	Google Scholar	16	No	Yes	10.1136/rmdopen-2023-004032
6	Gujarati translation, validity and reliability of Walking Impairment Questionnaire in people with intermittent claudication due to peripheral artery disease	Journal of Vascular Nursing	2024	1.1	1	Google Scholar	5	Senior	Yes	10.1016/j.jvn.2023.11.003
7	Global, regional, and national burden of osteoarthritis, 1990–2020 and projections to 2050: a systematic analysis for the Global Burden of Disease Study 2021	Lancet Rheumatology	2023	15.0	79	Google Scholar	251	No	Yes	10.1016/S2665-9913(23)00163-7
8	Global burden of peripheral artery disease and its risk factors, 1990–2019: a systematic analysis for the Global Burden of Disease Study 2019	Lancet Global Health	2023	19.9	35	Google Scholar	194	No	Yes	10.1016/S2214-109X(23)00355-8

9	Effect of a home-based, walking exercise behavior change intervention vs usual care on walking in adults with peripheral artery disease: the MOSAIC randomized clinical trial	JAMA	2022	63.1	48	Google Scholar	14	First	Yes	10.1001/jama.2022.3391
10	Gait rehabilitation for foot and ankle impairments in early rheumatoid arthritis: a feasibility study of a new gait rehabilitation programme (GREAT Strides)	Pilot and feasibility studies	2022	1.5	2	Google Scholar	17	Second	Yes	10.1186/s40814-022-01061-9
11	Inequity in exercise-based interventions for adults with intermittent claudication due to peripheral arterial disease: a systematic review	Disability & Rehabilitation	2022	2.5	4	Google Scholar	4	First	No	10.1080/09638288.2022.2102255
12	Using intervention mapping in the systematic development of a behaviour change intervention to enhance exercise adherence among people with persistent musculoskeletal pain	Physiotherapy Canada	2022	0.9	4	Google Scholar	3	Second	Yes	10.3138/ptc-2020-0083
13	“It’s important to buy in to the new lifestyle”: barriers and facilitators of exercise adherence in a population with persistent musculoskeletal pain	Disability & Rehabilitation	2021	2.5	33	Google Scholar	3	Second	Yes	10.1080/09638288.2019.1629700

14	Pre-operative prognostic factors for walking capacity after surgery for lumbar spinal stenosis: a systematic review	Age and Aging	2021	6.7	2	Google Scholar	8	Senior	Yes	10.1093/ageing/afab150
15	Effectiveness of web-based and mobile health interventions designed to enhance adherence to physical activity for people with inflammatory arthritis: a systematic review	Rheumatology advances in practice	2021	3.1	10	Google Scholar	9	Senior	Yes	10.1093/rap/rkab016
16	Smartphone apps targeting physical activity in people with rheumatoid arthritis: systematic quality appraisal and content analysis	JMIR mHealth and uHealth	2020	7.08	24	Google Scholar	8	First	Yes	10.2196/18495
17	Motivating Structured walking Activity in people with Intermittent Claudication (MOSAIC): protocol for a randomised controlled trial of a physiotherapist-led, behavioural change intervention versus usual care in adults with intermittent claudication	BMJ Open	2019	2.9	17	Google Scholar	11	First	Yes	10.1136/bmjopen-2019-030002

18	Are Walking Treatment Beliefs and Illness Perceptions Associated With Walking Intention and 6-Min Walk Distance in People With Intermittent Claudication? A Cross-Sectional Study	Journal of Aging & Physical Activity	2019	1.4	10	Google Scholar	3	Senior	Yes	10.1123/japa.2018-0245
19	A randomized controlled feasibility trial of a home-based walking behavior–change intervention for people with intermittent claudication	Journal of Vascular Nursing	2019	1.1	23	Google Scholar	3	Senior	Yes	/10.1016/j.jvn.2018.11.001
20	Behaviour change techniques associated with adherence to prescribed exercise in patients with persistent musculoskeletal pain: systematic review	British Journal of Health Psychology	2019	5.2	128		5	Second	yes	10.1111/bjhp.12324

Please list below other significant publications, outputs or products (see guidance notes).

Book Chapters

Bearne L.M, Opava C. H. Physical Activity and Exercise in Rheumatoid Arthritis In: The Oxford Textbook of Rheumatoid Arthritis Eds: Scott D.L, Galloway J, Cope A, Pratt. A, Strand V. 2021 Oxford University Press.

Dziedzic K, Bearne L.M. The principles of Physiotherapy In: Musculoskeletal Disorders in Primary Care Ed: Warburton L. 2012 Royal College of Physicians ISBN: 978-0-850-84333-0

Hurley M.V, Bearne L.M. The principles of therapeutic exercise and physical activity. In: Rheumatology. Evidence based Practice for Physiotherapists and Occupational Therapists. Eds: Dziedzic K, Hammond A, 2010 Elsevier Ltd, UK. IBSN 978-0-443-06934-5

Bearne L.M, Hurley M.V. Physical therapies - treatment options in rheumatology. In: Rheumatology. Evidence based Practice for Physiotherapists and Occupational Therapists Eds: Dziedzic K, Hammond A, 2010 Elsevier Ltd, UK. IBSN 978-0-443-06934-5

Bearne L.M, Hurley M.V. Exercise in arthritis?!: Questions, answers and advice for patients and practitioners. In: Nutrition and Rheumatic Diseases. Ed: Coleman L. 2008 The Humana Press Incorporated, USA. ISBN 978-1-58829-976-5

Reports

Sekhon M, Whitney J, Godfrey E, Bearne L. M. The experiences of and views on rehabilitation needs of adults with long term COVID-19: Report from patient and healthcare professional stakeholder engagement 2021 King's College Hospital Charity

Hurley M.V, Dzeidic K, Bearne L.M, Sim J. Clinical and cost effectiveness of physiotherapy in the management of elderly people with rheumatological conditions. 2001 Evidenced-based report for the Chartered Society of Physiotherapy

Hurley M.V, Bearne L.M, Scott D.L. Effective and safe rehabilitation of patients with rheumatoid arthritis. 2000 NHS National Research and Development Programme

Guidance/clinical recommendations

Chapman L, Backhouse M, Corp N, van der Windt D, Bearne L , Cherry L, Cleary G, Davey J, Ferguson R, Helliwell P, Lomax A, McKeeman H, Rawlings A, Rees R, Rooney R, Ryan S, Sanders L, Siddle H, Varley S, Warburton L, Woodburn J, Roddy E, for the British Society for Rheumatology Guideline Steering Group The British Society for Rheumatology guideline for the management of foot health in inflammatory arthritis. Under consultation

Sekhon, M., de Thurah, A., Fragoulis, G.E., Schoones, J., Stamm, T.A., Vlieland, T.P.V., Esbensen, B.A., Lempp, H., Bearne, L., Kouloumas, M. and Pchelnikova, P., 2024. Synthesis of guidance available for assessing methodological quality and grading of evidence from qualitative research to inform clinical recommendations: a systematic literature review. RMD open, 10(2), p.e004032.